L^2 HODGE THEORY AND THE LLV ALGEBRA OF SINGULAR K3 MODULI SPACES

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Abstract. We identify the Looijenga–Lunts–Verbitsky algebra on the intersection cohomology of singular moduli spaces of sheaves on K3 surfaces with the orthogonal Lie algebra of their second cohomology enlarged by a hyperbolic plane. The surfaces have Picard rank one, the Mukai vector is twice a primitive vector of square at least four, and the rank and polarisation satisfy numerical conditions ensuring that all semistable sheaves are locally free. These moduli spaces have two singular strata and admit no symplectic resolution. For their natural hyperkähler metric, we identify square integrable cohomology with intersection cohomology and prove that harmonic forms are preserved by the three Lefschetz operators and their adjoints. The analytic argument extends the Kähler cone theorem to singular links and establishes invariance of operator domains under metric comparisons that distinguish tangential and normal directions at the strata. Monodromy then extends the relations supplied by one positive three-plane to the full second cohomology. We also determine the subrepresentation generated by the unit and, under an explicit Hodge-theoretic comparison hypothesis, the complete representation in dimension eighteen.

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1. Introduction

The Lefschetz operators of a compact hyperkähler manifold generate much more than the $\mathfrak{sl}_2$ associated with one Kähler class. Together with their lowering operators they generate the orthogonal Lie algebra of the Beauville–Bogomolov space enlarged by a hyperbolic plane

Date: November 14, 2026
This version: November 14, 2026.
2020 *Mathematics Subject Classification.* 14J42 (primary); 14J60, 14F43, 53C26, 58A14 (secondary).
Key words and phrases. K3 surfaces, moduli of sheaves, intersection cohomology, L^2 Hodge theory, hyperkähler quotients, Looijenga–Lunts–Verbitsky algebra.

[Ver96, LL97]. For a singular symplectic variety, intersection cohomology is the natural setting for the same question. Hard Lefschetz is available there, but an analytic proof of the additional commutation relations must also specify the domains of the differential operators near the singularities.

We treat a family of singular moduli spaces on K3 surfaces for which the regular locus carries a natural hyperkähler metric. The singularities have positive-dimensional strata and no symplectic resolution. The main analytic task is to show that the pointwise hyperkähler identities act on square integrable harmonic forms. The algebraic task is then to recover all Lefschetz operators from the three Kähler forms of this one metric. We separate these two arguments: a general algebraic criterion appears in Section 2, followed by the analysis which verifies it for the moduli spaces.

1.1. **The moduli spaces.** Let S be a projective K3 surface with $\mathrm{Pic}(S) = \mathbb{Z}A$, where A is ample. Fix a primitive Mukai vector $v_0 = (r_0, \ell A, s_0)$ such that

$$v_0^2 = 2k, \qquad k \geq 2, \qquad r_0 \geq 2k+1, \qquad \gcd(r_0, \ell A^2) = 1. \tag{1}$$

Write $M = M_A(2v_0)$ and $M_0 = M_A(v_0)$ for the moduli spaces of A-Gieseker-semistable sheaves. Under these assumptions the sheaves are locally free and their slope and Gieseker stability notions agree as required in Section 3. The singular locus and its diagonal are

$$\Sigma = \mathrm{Sym}^2 M_0 \subset M, \qquad \Delta = \{[F \oplus F] : [F] \in M_0\} \cong M_0.$$

Their complex dimensions are

$$\dim M = 8k+2, \qquad \dim \Sigma = 4k+4, \qquad \dim \Delta = 2k+2.$$

The space M is an irreducible symplectic variety with $b_2(M) = 23$ [PR24, PR23]. Its Beauville–Bogomolov form q_M has real signature $(3, 20)$. It is locally factorial in the Zariski topology and terminal, and has neither a symplectic resolution nor a smoothing [KLS06, Nam01, Nam06]. The qualification on the topology matters: the transverse analytic singularity at a point of $\Sigma \setminus \Delta$ is not analytically $\mathbb{Q}$-factorial.

Put $2n = \dim_{\mathbb{C}} M$, so $n = 4k+1$. Intersection cohomology is graded in the usual cohomological degrees. A class $x \in H^2(M;\mathbb{Q})$ acts on it by L_x. We say that x is a *Lefschetz class* if

$$L_x^j : IH^{2n-j}(M;\mathbb{Q}) \longrightarrow IH^{2n+j}(M;\mathbb{Q})$$

is an isomorphism for every $j \geq 0$. For such a class there is a unique operator Λ_x of degree -2 satisfying $[L_x, \Lambda_x] = h$, where h acts in degree j by $j - 2n$. The Lie algebra generated by these triples is the *Looijenga–Lunts–Verbitsky algebra*, abbreviated LLV algebra.

Theorem 1.1. *For M satisfying (1), the Lefschetz classes in $H^2(M;\mathbb{Q})$ are exactly the classes with $q_M(x) \neq 0$, and*

$$\mathfrak{g}(M;\mathbb{Q}) \cong \mathfrak{so}\big((H^2(M;\mathbb{Q}), q_M) \oplus U\big),$$

where U is a hyperbolic plane. Its real form is $\mathfrak{so}(4,21)$, and its dimension is 300. The submodule generated by $1 \in IH^0(M;\mathbb{C})$ is the irreducible representation with highest weight $(n, 0, \ldots, 0)$ and occurs once. In particular,

$$\mathrm{Sym}^j H^2(M;\mathbb{C}) \hookrightarrow IH^{2j}(M;\mathbb{C}), \qquad \dim IH^{2j}(M) \geq \binom{22+j}{j} \quad (0 \leq j \leq n).$$

Here and below a highest weight is written in orthogonal coordinates for the root system B_{12}. The first example is a quartic K3 surface of Picard rank one with $v_0 = (5, A, 0)$; then $k = 2$ and $\dim_{\mathbb{C}} M = 18$.

The LLV theorem is known for varieties admitting a symplectic resolution [FSY22], and for the isolated-singularity and orbifold cases treated by Tighe [Tig25]. The spaces above are outside those classes. Their terminalisation is the space itself, and the transverse singularity along $\Sigma \setminus \Delta$ is the closure of the minimal nilpotent orbit in $\mathfrak{sl}_{2k}$. Its resolution $T^*\mathbb{P}^{2k-1}$ is small, so this germ is not analytically $\mathbb{Q}$-factorial and cannot be a finite quotient singularity.

1.2. **The analytic statement.** The metric used throughout is the L^2 metric on $M_{\mathrm{reg}} = M \setminus \Sigma$ obtained from Hermitian–Einstein connections on S. Its existence, its Kähler class and the comparison with flat hyperkähler quotients are imported from [Wue26a, Wue26b]; the precise input is stated in Recollection 3.1. In particular, the metric comparison imported there is a bound on the metric itself, without a derivative bound. The first-order estimates needed below are proved separately.

Let I_a, ω_a, $a = 1, 2, 3$, denote the parallel complex structures and Kähler forms of this metric. On differential forms let W_a be the Weil operator, acting on (p, q)-forms for I_a by i^{p-q}, and put $d_a^c = W_a^{-1} d W_a$. All closed extensions below are taken in the L^2 space of the regular locus.

Theorem 1.2. *For the metric g on M_{reg}:*

(1) *$d_{\min} = d_{\max}$, and the maximal de Rham complex has finite-dimensional cohomology and closed range. There is a canonical isomorphism*

$$IH^\bullet(M;\mathbb{R}) \cong H^\bullet_{(2)}(M_{\mathrm{reg}}, g) \cong \mathcal{H}^\bullet, \qquad \mathcal{H} = \ker d_{\max} \cap \ker \delta_{\max}.$$

(2) *The isomorphism intertwines the $H^\bullet(M;\mathbb{R})$ action with wedge product by bounded closed representatives. It identifies the intersection pairing with integration of wedge products.*

(3) *Each W_a preserves the Gaffney domain $V = \operatorname{Dom} d_{\max} \cap \operatorname{Dom} \delta_{\max}$. Each harmonic form is d_a^c-closed and d_a^c-coclosed. Wedge product by ω_a and its pointwise adjoint preserve $\mathcal{H}$ and satisfy Verbitsky's $\mathfrak{so}(4,1)$ relations there.*

The first assertion follows from local cone cohomology and the characterisation of the intersection complex. The third is more sensitive to the metric. Equality of the formal Laplacians for d and d_a^c on the regular locus does not identify their closed extensions. It is the invariance of V that permits the Kähler identities to be used on harmonic forms.

Two features of the proof are useful beyond this family. First, the Kähler cone theorem is extended from smooth links to singular links with discrete Hodge spectrum and the L^2 Stokes property. The potentially problematic modes are homogeneous closed and coclosed forms at the critical indicial exponents. The Kähler identities force their d^c derivatives into gaps in the exponents of adjacent degrees; an integration-by-parts argument on the link treats the endpoint. This is the mechanism of Cheeger's smooth-link theorem [Che83a], arranged here to permit induction on depth.

Second, the comparison of Gaffney domains keeps track of directions. At a single stratum, a metric error of order the distance to that stratum, with bounded first derivative, is sufficient. Near the diagonal there are two distances. Errors in the normal and mixed directions must vanish at the nearer stratum, whereas tangential errors may be of the larger order. Tangential regularity on a product model controls the latter. These estimates are verified using finite-dimensional slices of the gauge-theoretic quotient.

1.3. **From one metric to all classes.** Theorem 1.2 provides Lefschetz triples only on the three-plane $\mathcal{P} = \langle [\omega_1], [\omega_2], [\omega_3] \rangle$. The Beauville–Bogomolov form is a positive multiple of the Euclidean form on this plane. The monodromy of locally trivial deformations has Zariski-dense image in the special orthogonal group of H^2. The operator relations needed for the LLV theorem are algebraic in at most three classes. Every nondegenerate three-plane is an orthogonal translate of $\mathcal{P}_{\mathbb{C}}$; monodromy invariance therefore extends those relations to all classes. This proves Theorem 1.1 without requiring an analytic theorem for every Kähler class.

Section 9 also shows that a subgroup of finite index of monodromy acts through the spin group of H^2. The intersection pairing and the positive L^2 inner product make the possible centralising factor finite. Section 10 applies this to the eighteen-dimensional example. Its complete decomposition requires a specified isomorphism of graded rational Hodge structures motivated by the BPS and χ-independence results of [DHS22, DHKSV26, GWZ26]. We state that premise explicitly; a calculation of graded dimensions alone does not identify an LLV representation.

The analytic background is the L^2 theory of Cheeger and Cheeger–Goresky–MacPherson [Che80, CGM82], its Hilbert complex formulation [BrL92, BrL93], and the edge theories of [HM05, ALMP12, ALMP18]. Our metric is not assumed to have an iterated edge normal form. The quotient stratification is consistent with [DS97, May22]; the charts used here additionally control the metric and all three complex structures. No identification of the harmonic type decomposition with Saito's Hodge structure is used in the LLV theorem or in the conditional representation argument.

2. The algebra determined by one three-plane

We first isolate the algebraic implication of the analytic theorem. Let X be a projective primitive symplectic variety of complex dimension $2n$, with $b_2(X) \geq 5$ and an injection $H^2(X;\mathbb{Q}) \to IH^2(X;\mathbb{Q})$. Set

$$H = H^2(X;\mathbb{C}), \qquad V_X = IH^\bullet(X;\mathbb{C}), \qquad q = q_X.$$

The letter H in this section denotes cohomology; in the quotient constructions below it will denote a compact stabiliser. We retain the grading operator $h|_{IH^j} = j - 2n$ and the notation L_x, Λ_x from the introduction.

Theorem 2.1. *Suppose that three real Lefschetz classes x_1, x_2, x_3 satisfy $q(x_a, x_b) = c\delta_{ab}$ with $c > 0$. Put $L_a = L_{x_a}$, $\Lambda_a = \Lambda_{x_a}$ and $K_{ab} = [L_a, \Lambda_b]$ for $a \neq b$. Assume*

$$[\Lambda_a, \Lambda_b] = 0, \qquad K_{ab} = -K_{ba}, \qquad [K_{ab}, L_c] = 2(\delta_{bc} L_a - \delta_{ac} L_b). \tag{2}$$

Then the LLV algebra of X is $\mathfrak{so}((H,q) \oplus U)$, with graded pieces

$$\mathfrak{g}_2 = L(H), \qquad \mathfrak{g}_0 \cong \mathfrak{so}(H,q) \oplus \mathbb{C}h, \qquad \mathfrak{g}_{-2} \cong H.$$

The isomorphism is defined over $\mathbb{Q}$. Its real form is $\mathfrak{so}(4, b_2(X) - 2)$.

2.1. **The geometric inputs to the criterion.** We use three established facts. Locally trivial deformations give parallel transport on ordinary and intersection cohomology, compatible with their module structure. Indeed these families are locally real-analytic fibre bundles [AV21, Proposition 5.1], and intersection cohomology and its ordinary-cohomology action are topological invariants [GM83]. Consequently their monodromy group satisfies

$$\gamma L_x \gamma^{-1} = L_{\gamma x}. \tag{3}$$

The image on H^2 has finite index in the integral orthogonal group [BL22, Theorem 1.1]; for the moduli spaces under consideration see also [OPR24]. Borel density implies that its complex Zariski closure contains $\mathrm{SO}(H, q)$.

Next, the Lefschetz locus is

$$H_{\mathrm{Lef}} = \{x \in H : q(x) \neq 0\}. \tag{4}$$

For rational classes this is [Tig25, Proposition 2.11]. To pass to complex classes, the non-Lefschetz locus is closed, conical and monodromy invariant, since it is defined by determinants of the maps L_x^j. It is therefore $\mathrm{SO}(H, q)$-invariant. The nonisotropic vectors form one orbit up to scaling, and this orbit contains an ample class. The nonzero isotropic vectors form one orbit. This orbit contains a rational point by Meyer's theorem, because the rational form is indefinite and has rank at least five. The rational statement proves (4).

Finally $\mathfrak{g}$ is semisimple. This is the polarised Lefschetz module criterion of [LL97, Proposition 1.6]: Saito's pure Hodge structures on IH, the Hodge-compatible action of H^2, and the Hodge–Riemann relations for an ample class supply its hypotheses [Sai88, Sai90, dCM05]. The intersection pairing is taken with the degree signs of [LL97, §1.9], so that the raising operators are skew. These facts concern intersection cohomology algebraically and make no assertion about types of harmonic forms.

2.2. **Extending the relations.** For a Lefschetz class the equations defining Λ_x have a unique solution. Solving this finite linear system shows that $x \mapsto \Lambda_x$ is a regular rational map on H_{Lef} [LL97, §1.1]. Uniqueness also gives $\Lambda_{tx} = t^{-1}\Lambda_x$ and the conjugacy analogue of (3).

Lemma 2.2. *Let $N \leq 3$ and let $\mathcal{D} \subset H^N$ be a dense invariant Zariski open set. A relatively closed monodromy-invariant subset of $\mathcal{D}$ which contains $\mathcal{D} \cap \mathcal{P}_{\mathbb{C}}^N$, where $\mathcal{P} = \langle x_1, x_2, x_3 \rangle$, is all of $\mathcal{D}$.*

Proof. The stabiliser of a closed algebraic subset is Zariski closed, so the subset is invariant under $\mathrm{SO}(H, q)$. Tuples with independent entries and nondegenerate span are dense. Each such span is contained in a nondegenerate three-plane. Witt's extension theorem carries $\mathcal{P}_{\mathbb{C}}$ to this plane by an orthogonal transformation. A reflection in its orthogonal complement corrects the determinant if necessary; that complement has positive dimension. Invariance therefore places every such tuple in the closed subset, proving the claim by density. □

For $x = \sum_a t_a x_a \in \mathcal{P}_{\mathbb{C}}$ the first two relations in (2) imply

$$[\sum_a t_a L_a, \sum_a t_a \Lambda_a] = (\sum_a t_a^2)h.$$

Uniqueness of the lowering operator gives

$$q(x)\Lambda_x = c \sum_a t_a \Lambda_a. \tag{5}$$

Thus lowering operators commute on this plane. Their commutator vanishes everywhere on H^2_{Lef} by Lemma 2.2.

Define initially $A_x = q(x)\Lambda_x$ for $q(x) \neq 0$. Equation (5) gives additivity on pairs in $\mathcal{P}_{\mathbb{C}}$ for which $x, y, x + y$ are nonisotropic. Lemma 2.2 extends it to every such pair. Together with homogeneity this extends A to a linear map on H. For clarity, one may differentiate $A(x + y) = A(x) + A(y)$ at generic pairs: $dA_{x+y} = dA_x = dA_y$, so dA is constant on its irreducible domain; homogeneity removes the constant term.

Use the convention $(x \wedge y)z = q(y,z)x - q(x,z)y$ for $\mathfrak{so}(H,q)$. The last relation of (2), together with $[h, L_z] = 2L_z$, now gives on the three-plane

$$[[L_x, A_y], L_z] = 2L_{q(x,y)z+(x\wedge y)z}.$$

Both sides are regular for $q(y) \neq 0$ before the linear extension of A. Lemma 2.2 with three entries, followed by linearity, therefore proves

$$[A_x, A_y] = 0, \qquad [[L_x, A_y], L_z] = 2L_{q(x,y)z+(x\wedge y)z} \quad (x,y,z \in H). \tag{6}$$

2.3. **Recovering the orthogonal algebra.**

Proof of Theorem 2.1. The map $L : H \to \mathrm{End}(V_X)$ is injective, since $L_x 1$ is the image of x in IH^2. Let $\mathfrak{k}$ be generated by the brackets $[L_x, A_y]$. Equation (6) defines a Lie homomorphism $\pi : \mathfrak{k} \to \mathfrak{gl}(H)$ by

$$[T, L_z] = L_{\pi(T)z}, \qquad \pi([L_x, A_y]) = 2q(x,y)\mathrm{id} + 2x \wedge y.$$

For $T \in \mathfrak{k}$, conjugation by e^{sT} preserves h and takes L_z to $L_{e^{s\pi(T)}z}$. It takes the corresponding lowering operator to the corresponding lowering operator. Hence

$$[T, \Lambda_z] = \frac{A_{\pi(T)z}}{q(z)} - \frac{2q(z, \pi(T)z)}{q(z)^2} A_z. \tag{7}$$

It follows that $L(H) \oplus \mathfrak{k} \oplus A(H)$ is closed under brackets and equals $\mathfrak{g}$. Its three summands have degrees $2, 0, -2$. Conjugation by the Weyl element of any Lefschetz $\mathfrak{sl}_2$ exchanges the two extreme pieces, so A is also injective.

The image of π is $\mathfrak{so}(H,q) \oplus \mathbb{C}\mathrm{id}$, and $\pi(h) = 2\mathrm{id}$. If $\pi(T) = 0$, then T commutes with all raising operators and, by (7), all lowering operators. It is central in the semisimple algebra $\mathfrak{g}$, and is zero. Thus π is an isomorphism. If $\pi(T) = a\mathrm{id} + B$ with $B \in \mathfrak{so}(H,q)$, the remaining brackets are

$$[T, L_z] = L_{az+Bz}, \qquad [T, A_z] = A_{-az+Bz}, \qquad [L_x, A_y] = \pi^{-1}(2q(x,y)\mathrm{id} + 2x \wedge y).$$

Choose isotropic generators u_+, u_- of U with $\widetilde{q}(u_+, u_-) = 1$. All the brackets agree under

$$L_x \longmapsto u_+ \wedge x, \quad A_y \longmapsto -2u_- \wedge y, \quad \pi^{-1}(B) \longmapsto B, \quad h \longmapsto 2u_+ \wedge u_-. \tag{8}$$

For example $[u_+ \wedge x, u_- \wedge y] = -x \wedge y - q(x,y)u_+ \wedge u_-$. The indicated elements exhaust the orthogonal algebra and have the same graded dimensions. This proves the complex isomorphism.

The maps L, A and π are rational: their values on rational nonisotropic classes are rational, and those classes are Zariski dense. Formula (8) is therefore defined over $\mathbb{Q}$. The real signature of $q \oplus U$ is $(4, b_2 - 2)$. □

2.4. **Application to the moduli space.** For the moment use Theorem 1.2, proved below. The bounded Kähler forms represent classes in $H^2(M;\mathbb{R})$. Proposition 8.3 proves $q_M([\omega_a], [\omega_b]) = c\delta_{ab}$ with $c > 0$. Under the harmonic representative isomorphism, $L_{[\omega_a]}$ is pointwise wedge product and its lowering operator is the pointwise adjoint. Indeed these operators preserve harmonic forms and constitute an $\mathfrak{sl}_2$-triple. Finite-dimensional $\mathfrak{sl}_2$ theory gives hard Lefschetz, and uniqueness then identifies the lowering operator. The pointwise hyperkähler relations are (2). Also $H^2(M;\mathbb{Q}) \cong IH^2(M;\mathbb{Q})$ by [Tig25, Proposition 2.17]. Theorem 2.1 proves the algebra statement in Theorem 1.1; its dimension is $\binom{25}{2} = 300$.

To identify the unit submodule, the negative part of $\mathfrak{g}$ annihilates 1, the semisimple part of $\mathfrak{g}_0$ acts trivially on it, and $h1 = -2n$. Complete reducibility shows that a module generated

by this lowest-weight vector is the irreducible $\mathbb{V}_{(n)}$. The raising operators generate the same module, and the usual description [LL97, Proposition 2.14] is

$$\mathbb{V}_{(n)} \cong \operatorname{Sym}^\bullet H^2 / \langle x^{n+1} : q_M(x) = 0 \rangle. \tag{9}$$

There can be only one such summand because $IH^0(M) = \mathbb{C}$. The ideal has no element of degree at most n, giving the injections and dimension bounds in Theorem 1.1.

3. The metric and its local models

We record the geometric input before analysing differential forms. The Hermitian–Einstein connections below are taken with respect to the Ricci-flat Kähler metric in A on S. The determinant is fixed, and gauge transformations are divided by the scalar action.

Recollection 3.1 ([Wue26a, Wue26b]). Under (1), the following statements hold.

(1) Every semistable sheaf of vector $2v_0$ is locally free. Each point of M has a unique slope-polystable representative: a stable bundle, a sum $E_1 \oplus E_2$ of nonisomorphic stable bundles of vector v_0, or $F \oplus F$. These are respectively the strata M_{reg}, $\Sigma \setminus \Delta$ and Δ.
(2) The gauge-theoretic hyperkähler quotient is homeomorphic to M. Its regular-locus metric, with a fixed normalisation, is the L^2 metric g on harmonic representatives. Its first Kähler form represents a positive multiple of an ample determinant class, and $\omega_2 + i\omega_3$ is a nonzero multiple of the holomorphic symplectic form.
(3) At a polystable bundle E, put $\mathbb{E} = H^1(S, \operatorname{End} E)$ and let H be its unitary automorphism group modulo scalars. For the flat hyperkähler quotient $\mathbb{E} /\!\!/\!\!/_0 H$, with metric g_0 and radius r, there is a biholomorphism of germs Ψ_E onto $(M, [E])$ such that

$$(1 - C_E r^2) g_0 \le \Psi_E^* g \le (1 + C_E r^2) g_0. \tag{10}$$

No estimate for derivatives is included in this assertion.
(4) The metric completion of (M_{reg}, g) is M.

The bundle statement is [Wue26b, Proposition 2.1]; the quotient, metric and chart statements are [Wue26a, Theorem 1.1] and [Wue26b, Theorem 1.1, Lemma 5.2 and §5.3]; completion is [Wue26b, Proposition 4.2]. The holomorphic quotient model is provided by [AS18, BZ19]. The hyperkähler structure is the gauge-theoretic one of [HKLR87, Ito88], using [Don85, UY86]. These are imported results. In particular, the analytic conclusions of this paper retain their dependence on the companion metric theorems, which are preprints.

3.1. **The two transverse cones.** At $E = E_1 \oplus E_2$ with $E_1 \not\cong E_2$, stability and Serre duality give $\operatorname{Hom}(E_1, E_2) = \operatorname{Ext}^2(E_1, E_2) = 0$. Riemann–Roch then gives $\dim_{\mathbb{C}} \operatorname{Ext}^1(E_1, E_2) = 2k$. The effective stabiliser is a circle, and its slice decomposes as

$$\mathbb{E} = \mathbb{E}^H \oplus \mathbb{E}_1, \qquad \mathbb{E}^H \cong \mathbb{C}^{4k+4}, \qquad \mathbb{E}_1 \cong T^*\mathbb{C}^{2k}.$$

The circle has weights $1, -1$ on the two summands. Its punctured zero fibre is connected and the action on it is free. Thus

$$C_1 = C(Z_1) = T^*\mathbb{C}^{2k} /\!\!/\!\!/_0 \mathrm{U}(1) \tag{11}$$

has smooth compact link of real dimension $8k - 5$. The map $(z, \zeta) \mapsto z \otimes \zeta$ identifies its complex analytic space with the closure of the minimal nilpotent orbit of $\mathfrak{sl}_{2k}$.

At $E = F \otimes \mathbb{C}^2$, the effective group is $PU(2) = \mathrm{SO}(3)$. The real harmonic deformation space of F is $\mathbb{H}^{k+1}$, and the decomposition of Hermitian 2×2 matrices into scalar and traceless parts gives

$$\mathbb{E} = \mathbb{H}^{k+1} \oplus (\mathbb{R}^3 \otimes \mathbb{H}^{k+1}), \qquad C_2 = C(Z_2) = (\mathbb{R}^3 \otimes \mathbb{H}^{k+1}) /\!\!/_0 \, \mathrm{SO}(3). \tag{12}$$

The group acts on the first factor of the tensor product. Regard $n \in \mathbb{R}^3 \otimes \mathbb{H}^{k+1}$ as a map from $\mathbb{R}^3$ to $\mathbb{H}^{k+1}$. Its stabiliser is trivial in rank at least two, a circle in rank one, and $\mathrm{SO}(3)$ at zero. Rank-one tensors belong to the zero fibre. Their quotient is $(\mathbb{H}^{k+1} \setminus \{0\})/\{\pm 1\}$, the only singular stratum of $C_2 \setminus \{0\}$.

At $n = e \otimes l$ the slice to the quaternionic orbit directions is

$$(e \otimes \mathbb{H}^{k+1}) \oplus \big(e^\perp \otimes (\mathbb{H}l)^\perp\big). \tag{13}$$

The stabilising circle fixes the first summand and acts on the second as on $T^*\mathbb{C}^{2k}$. The transverse cone is therefore C_1. The link Z_2 has real dimension $12k-1$ and singular locus $\mathbb{RP}^{4k+3}$, of codimension $8k-4$.

3.2. **Topology and the two distances.** The quotient charts give a stratified pseudomanifold with filtration $M \supset \Sigma \supset \Delta$. Here the real codimensions are $8k-4$ and $12k$, both even. To see the local product description on Z_2, choose a rank-one direction $e \otimes l_0$. Near it the largest singular value has a unique eigenline. Moving this line to $\mathbb{R}e$ writes a tensor as $e \otimes l + f$, where the zero moment map equations impose $f \in e^\perp \otimes (\mathbb{H}l)^\perp$ and the circle moment map equation on f. A quaternionic trivialisation of $(\mathbb{H}l)^\perp$ gives local coordinates $(l, [f])$, with

$$r^2 = |l|^2 + |f|^2.$$

On $r = 1$, the coordinates $(l/|l|, [f]/|l|)$ identify a neighbourhood with a ball in the rank-one stratum times a truncated C_1. This also proves the claimed link description. The biholomorphic charts Ψ_E preserve the filtration because they preserve the singular locus and then its singular locus.

Two distances will enter the analysis near Δ:

$$r = \mathrm{dist}(\,\cdot\,, \text{vertex stratum}), \qquad \rho = \mathrm{dist}(\,\cdot\,, \text{entire singular set}).$$

They are comparable away from the rank-one directions but need not be comparable near them. Estimate (10) controls L^2 de Rham cohomology, whose domains are invariant under quasi-isometry. To control $d+\delta$ we will construct different charts, generally nonholomorphic, with first-order agreement in the appropriate directions.

4. Domains on Kähler cones

4.1. **The domain condition.** A singular Riemannian space will mean a locally compact metric space X with dense smooth regular locus, whose Riemannian length metric is the restriction of its distance. Forms are complex valued and are defined on X_{reg}. Their support is taken in X, so compact support may meet the singular set. The maximal operators d and δ are defined by distributional differentiation on X_{reg}; the minimal operator $d_{\min}$ is the graph closure of d on smooth forms compactly supported in X_{reg}.

We say that the L^2 *Stokes property* holds when $d_{\min} = d_{\max}$. Then $\delta_{\max} = d^*_{\max}$, and Hilbert complex theory [BrL92] makes

$$D = d + \delta, \qquad V(X) = \mathrm{Dom}\, d_{\max} \cap \mathrm{Dom}\, \delta_{\max}$$

self-adjoint, with $\|Du\|^2 = \|du\|^2 + \|\delta u\|^2$. The domain $V(X)$ is called the Gaffney domain. It need not be the minimal domain of D. Bounded Lipschitz functions preserve all the domains just defined.

Suppose g is Kähler on the regular locus. Write L for wedge product by its Kähler form, $\Lambda = L^*$, W for the Weil operator, and $d^c = W^{-1}dW$, $\delta^c = W^{-1}\delta W$, $D^c = W^{-1}DW$. The identities on smooth forms are

$$[L, \delta] = d^c, \quad [\Lambda, d] = -\delta^c, \quad [L, d] = [\Lambda, \delta] = 0, \quad D^2 = (D^c)^2, \tag{14}$$

and each of d, δ anticommutes with each of d^c, δ^c.

Definition 4.1. The Gaffney domain is *Weil invariant* on $\Omega \subset X$ if $Wu \in V(X)$ for every $u \in V(X)$ compactly supported in Ω.

The condition is local by Lipschitz partitions of unity. It is equivalent to $d^c u, \delta^c u \in L^2$ for those forms. Since V and W preserve degree, it can be checked on pure-degree forms, where the two terms of $D^c u$ have different degrees. Assuming also the Stokes property, on a fixed compact support the closed graph theorem gives

$$\|D^c u\| \le C(\|Du\| + \|u\|). \tag{15}$$

Indeed the distributional operator D^c is closed from the corresponding graph-norm space to L^2.

Proposition 4.2. *If X is compact and satisfies the Stokes property and Weil invariance, then $d^c\eta = \delta^c\eta = 0$ for every $\eta \in \ker d \cap \ker \delta$. The operators W, L, Λ preserve this harmonic space.*

Proof. Weil invariance makes $d^c\eta$ and $\delta^c\eta$ square integrable. They are closed and coclosed by the anticommutation identities, hence belong to V. Applying Weil invariance again and integrating by parts gives

$$\|d^c\eta\|^2 + \|\delta^c\eta\|^2 = \langle \eta, (D^c)^2\eta\rangle = \langle \eta, D^2\eta\rangle = 0.$$

All terms in these integrations lie in the required maximal domains; the Stokes property supplies the adjoint relation. Equation (14) now proves the assertions for L and Λ, and the definition proves the assertion for W. □

On a hyperkähler regular locus the pointwise operators satisfy

$$\begin{gathered}[L_a, L_b] = [\Lambda_a, \Lambda_b] = 0, \quad [L_a, \Lambda_a] = h, \quad K_{ab} = -K_{ba}, \\ [K_{ab}, L_c] = 2(\delta_{bc}L_a - \delta_{ac}L_b), \qquad [K_{ab}, \Lambda_c] = 2(\delta_{bc}\Lambda_a - \delta_{ac}\Lambda_b).\end{gathered} \tag{16}$$

These are Verbitsky's $\mathfrak{so}(4,1)$ relations [Ver90, LL97]. Consequently, Stokes and Weil invariance for all three structures give this action on harmonic forms.

4.2. **Separation of variables.** Let Z be compact of odd real dimension f, and consider the complete cone at infinity with vertex added, $C(Z)$, with metric $dr^2 + r^2 g_Z$. Set $m = f + 1$. We require the Stokes property on Z and on $C(Z)$, and compact resolvent of D_Z on $V(Z)$. Such a link will be called *admissible*. A smooth closed odd-dimensional link is admissible. More generally, the Stokes property on the cone follows from the first two requirements and oddness of f by Cheeger's cone theorem; see [GL02, §2] for the precise Stokes criterion.

Compact resolvent gives an orthonormal eigenbasis on Z, consisting of harmonic forms and pairs

$$\delta_Z\alpha = 0, \quad \Delta_Z\alpha = \lambda\alpha, \quad \beta = \lambda^{-1/2}d_Z\alpha, \quad \delta_Z\beta = \sqrt{\lambda}\,\alpha \quad (\lambda > 0).$$

The nonzero spectrum is discrete with finite multiplicities. For degree j, write a cone form in unitary radial coordinates as

$$u = r^{j-f/2}\varphi(r) + dr \wedge r^{j-1-f/2}\psi(r) \equiv (\varphi;\psi).$$

Its square norm is $\int(\|\varphi\|_Z^2 + \|\psi\|_Z^2)\,dr$. Direct differentiation gives

$$d(\varphi;\psi) = \big(r^{-1}d_Z\varphi;\,(\partial_r + (j-f/2)/r)\varphi - r^{-1}d_Z\psi\big), \tag{17}$$

$$\delta(\varphi;\psi) = \big(r^{-1}\delta_Z\varphi + (-\partial_r + (j-1-f/2)/r)\psi;\,-r^{-1}\delta_Z\psi\big). \tag{18}$$

A homogeneous form has coefficient pair $r^s(a;b)$; it is L^2 at zero precisely when $s > -1/2$. The operators d, δ lower s by one, whereas W, L, Λ and the Hodge star preserve it.

Lemma 4.3. *The homogeneous closed and coclosed forms of degree j with L^2 link coefficients are sums of the following modes:*

(1) $r^{s_\pm}(b_\pm\beta;\alpha)$, *where* $\deg\alpha = j-1$,

$$s_\pm = -\tfrac{1}{2} \pm \sqrt{\lambda + (j-m/2)^2}, \qquad b_\pm = \frac{\sqrt{\lambda}}{s_\pm + j - f/2};$$

(2) $(r^{f/2-j}\eta;0)$ *for* $\eta \in \mathcal{H}^j(Z)$;
(3) $(0;r^{j-1-f/2}\eta)$ *for* $\eta \in \mathcal{H}^{j-1}(Z)$.

In degree $j \neq m/2$ there is no exponent in $(-1/2-|j-m/2|, -1/2+|j-m/2|)$.

Proof. Equations (17)–(18) require the tangential component to be closed and the radial component to be coclosed on Z. Their Hodge decompositions therefore leave only the displayed pairs and the harmonic modes. On a pair, with $\gamma = j - f/2$, the coefficient equations are

$$b' + \gamma b/r - \sqrt{\lambda}a/r = 0, \qquad \sqrt{\lambda}b/r - a' + (\gamma-1)a/r = 0.$$

The indicial equation is $(s+\gamma)(s-\gamma+1) = \lambda$. The harmonic modes solve the two remaining scalar equations. Every resulting exponent satisfies $|s+1/2| \geq |j-m/2|$. □

4.3. **The critical modes.** Suppose now the cone is Kähler for a dilation-invariant complex structure and that its Gaffney domain is Weil invariant away from the vertex. The exponents which can obstruct a cutoff at zero lie in $-1/2 < s \leq 1/2$.

Lemma 4.4. *Every homogeneous closed and coclosed form with exponent in $(-1/2, 1/2]$ and L^2 link coefficients is annihilated by both d^c and δ^c.*

Proof. Multiply the form u by a cutoff on an annulus. Weil invariance there shows that d^cu and δ^cu have L^2 link coefficients; their exponent is $s-1$. By Lemma 4.3, either u has degree $m/2$, or $s = 1/2$ and it is a harmonic-link mode of degree $m/2-1$ or $m/2+1$.

In the middle degree, $d^cu = -\delta Lu$ and $\delta^cu = d\Lambda u$ are closed and coclosed. They have degrees $m/2+1$ and $m/2-1$ and exponent in $(-3/2, -1/2]$, inside the excluded intervals of Lemma 4.3. They vanish.

For degree $m/2-1$, the same argument in degree $m/2-2$ first gives $\delta^cu = 0$. Set $\zeta = d^cu$. It is closed and coclosed and has degree $m/2$ and exponent $-1/2$. Also $d^c\zeta = \delta^c\zeta = 0$, the latter by $(D^c)^2u = D^2u = 0$. Hence $W\zeta = r^{-1/2}(\eta_1;\eta_2)$ with harmonic η_i on Z. Writing $Wu = r^{1/2}(a;b)$, the equations $d(Wu) = W\zeta$ and (17) give $\eta_1 = d_Za$, $\eta_2 = -d_Zb$. A harmonic form in the range of $d_{Z,\max}$ is zero by Stokes on Z. Thus $\zeta = 0$. Hodge star reduces degree $m/2+1$ to this case. □

Theorem 4.5 (The cone theorem)**.** *Let $C(Z)$ be a Kähler cone with admissible link. Weil invariance on the complement of the vertex implies Weil invariance at the vertex as well.*

For smooth links this is the cone result of [Che83a, Theorem 2.1]; see also [BrL93, Theorems 3.7 and 5.9]. The singular-link case uses the preceding lemma and the same scalar radial estimates. We include those estimates because no spectral gap at the critical exponents is assumed.

Decompose a pure-degree form into tangential-coexact/radial-exact terms, tangential-exact/radial-coexact terms, and harmonic-link terms. Testing weak equations against a link eigenform times a radial test function is legitimate by Stokes on the cone. It gives the same equations as (17)–(18). Truncations retain both slots of every coupled exact–coexact block: either slot alone can have a nonintegrable derivative which cancels only in the pair. The projections onto these complete blocks preserve V and are orthogonal both for the L^2 norm and the sum of the d and δ energies.

For the first type, the equations contain the separate terms $\sqrt{\lambda}a/r$ and $\sqrt{\lambda}b/r$, so

$$\|u_{\mathrm{ce}}/r\| \le \lambda_{\min}^{-1/2}\|Du\|. \tag{19}$$

For the second type the radial matrix is, up to a unitary sign,

$$\partial_r + r^{-1}\begin{pmatrix} \gamma & -\sqrt{\lambda} \\ -\sqrt{\lambda} & 1-\gamma \end{pmatrix}, \qquad \gamma = j - f/2.$$

Its eigenvalues are $\theta = 1/2 \pm \sqrt{\lambda + (j-m/2)^2}$. Harmonic-link terms give $\theta = \gamma$ or $1-\gamma$. Thus each coefficient satisfies

$$a' + \theta a/r = F. \tag{20}$$

The squares of the L^2 norms of the right sides sum to the corresponding part of $\|Du\|^2$.

Lemma 4.6. *Let $a, F \in L^2(0,R)$ satisfy (20), with $a = 0$ near R. Then:*

1. *for $\theta \ge 1/2$ or $\theta < -1/2$, $\|a/r\| \le |\theta + 1/2|^{-1}\|F\|$;*
2. *for $-1/2 < \theta < 1/2$, $a = c\chi r^{-\theta} + a_0$, where $\chi = 1$ near zero is a fixed cutoff and $|c| + \|a_0/r\| \le C_\theta(\|F\| + \|a\|)$;*
3. *for $\theta = -1/2$, $a = br^{1/2}$ and $\|b' r^{1/2}\| = \|F\|$.*

Proof. On $[\epsilon, R]$, integration by parts gives

$$\|a' + \theta a/r\|^2 = \|a' - a/(2r)\|^2 + (\theta + 1/2)^2\|a/r\|^2 + (\theta+1/2)[|a|^2/r]_\epsilon^R.$$

For $\theta \ge 1/2$, integrability forces $a(r) = r^{-\theta}\int_0^r t^\theta F(t)\,dt$, and the boundary term tends to zero. For $\theta < -1/2$ it is nonnegative. For the intermediate range the same integral formula includes the homogeneous term $cr^{-\theta}$; subtracting $c\chi r^{-\theta}$ leaves a vanishing boundary term and gives (2). The constant c is controlled on any fixed interior interval. The endpoint identity in (3) follows by differentiation. □

Proof of Theorem 4.5. Estimate (15) on a fixed annulus, dilation, and a dyadic partition of unity give

$$\|D^c w\| \le C(\|Dw\| + \|w/r\|) \tag{21}$$

for forms supported away from the vertex. In a fixed degree there are only finitely many $\theta \in [-1/2, 1/2)$, and the other values have $|\theta + 1/2|$ uniformly bounded away from zero. Equations (19) and Lemma 4.6 therefore decompose a compactly supported $u \in V$ as

$$u = w + \sum_{-1/2<\theta<1/2} c_\theta \chi u_\theta + \sum_{\theta=-1/2} b_\theta u_\theta,$$

where u_θ is a closed and coclosed mode of exponent $-\theta$, $w \in V$, and $\|Dw\| + \|w/r\| \leq C(\|Du\| + \|u\|)$. All sums are finite. Cut off w at $r = \epsilon$ with derivative bounded by C/r. Estimate (21) is uniform in ϵ, so distributional convergence gives $D^c w \in L^2$. Lemma 4.4 kills $D^c u_\theta$ in the finite sums. Their remaining derivative terms are square integrable by the cutoff estimate or by $\|b'_\theta r^{1/2}\| < \infty$. Thus $D^c u \in L^2$. Pure degree separates its two components, proving $Wu \in V$. □

Lemma 4.7. *The set of forms supported in $\{r < R\}$ with $\|u\| + \|Du\| \leq 1$ is precompact in $L^2(C(Z))$.*

Proof. Modes with link eigenvalue $\lambda > \lambda_0 \geq 4$ have total L^2 norm at most $CR/\sqrt{\lambda_0}$, by (19) and Lemma 4.6(1). There are finitely many remaining modes. Their coefficients have bounded $W^{1,2}$ norm on each $[\epsilon, R]$ and uniformly small mass near zero by Lemma 4.6. At its endpoint, $|b(r)| \leq \|b'\sqrt{r}\|\sqrt{\log(R/r)}$ since b vanishes near R. Rellich compactness on the interior intervals, followed by the uniform tail bounds, proves the assertion. □

5. Transferring the Gaffney domain

The domain of d depends only on the quasi-isometry class of a metric; that of δ does not. We now give the additional estimates which permit the cone theorem to be transferred to the moduli space. All statements about domains in an open set concern forms compactly supported in that set.

5.1. **Weighted interior regularity.** Put $\rho(x) = \operatorname{dist}(x, X_{\text{sing}})$. We require the following elementary form of bounded geometry: there are charts on balls of radius comparable to $\rho(x)$ in which the rescaled metric $\rho(x)^{-2}g$ is uniformly elliptic with uniformly bounded first derivatives, and the measure is locally doubling, with uniform constants. More precisely, charts from the Euclidean unit ball contain $\mathbb{B}(x, \kappa\rho(x))$ in their half-ball image and have image in $\mathbb{B}(x, \rho(x)/2)$, for fixed $\kappa > 0$. The bounds are understood on each region under discussion. Products of a compact smooth manifold with a smooth-link cone have this property.

Lemma 5.1. *Under these bounds, forms in V with support in a fixed compact set satisfy*

$$\|\rho\nabla u\| \leq C(\|\rho Du\| + \|u\|). \tag{22}$$

Proof. Interior elliptic regularity first gives $u \in W^{1,2}_{\text{loc}}$ on the regular locus. In a rescaled chart the principal coefficients of $d + \delta$ are uniformly elliptic and Lipschitz, and its zeroth-order coefficients are bounded. Freezing the principal coefficients on sufficiently small balls and using the Fourier estimate gives a uniform interior Gårding inequality. Rescaling back gives

$$\rho(x)^2 \int_{\mathbb{B}(x,\kappa\rho(x))} |\nabla u|^2 \leq C \int_{\text{chart}} (\rho(x)^2|Du|^2 + |u|^2).$$

Since ρ is Lipschitz, its values are comparable on such a chart. A maximal disjoint family of smaller balls has enlarged balls of bounded overlap by doubling. Summation proves (22). □

Definition 5.2. A chart from a model (X_0, g_0, I_0) to (X, g, I) is *adapted* if it is a homeomorphism, a diffeomorphism on regular loci, and its pullback (g', I') satisfies

$$cg_0 \leq g' \leq c^{-1}g_0, \quad |g' - g_0| + |I' - I_0| \leq C\rho, \quad |\nabla^0 g'| + |\nabla^0 I'| \leq C. \tag{23}$$

The model is Kähler on its regular part; all norms use g_0. An adapted metric chart requires only the metric bounds.

Theorem 5.3. *Suppose the model has the bounded geometry above. An adapted metric chart identifies its Gaffney domain with that of g', with equivalent graph norms on a fixed compact support. If the chart is adapted also for the complex structures, Weil invariance transfers from the model to g'.*

Proof. The same rescaled charts give bounded geometry for g', and the two connections differ by a bounded tensor. Hence (22) holds for both metrics with ∇^0. Let $S = *_0^{-1} *'$ on the graded exterior bundle. Then

$$\delta' = S^{-1}\delta_0 S, \qquad |S-1| \le C\rho, \qquad |\nabla^0 S| \le C.$$

For a local orthonormal frame e_i, differentiation gives

$$\delta_0(Su) - S\delta_0 u = -\sum_i \iota_{e_i}(\nabla^0_{e_i} S)u - \sum_i [\iota_{e_i}, S]\nabla^0_{e_i} u. \tag{24}$$

The right side is bounded by $C(|u| + \rho|\nabla^0 u|)$. The last sum is a first-order term and is essential. Estimate (22) controls it from either graph norm, proving both domain inclusions and their norm estimates.

For the complex structures put $Q = W_0^{-1}W'$. The Weil operator on degree j is the jth exterior power of the complex structure on covectors. Thus $|Q-1| \le C\rho$ and $|\nabla^0 Q| \le C$. The Leibniz formulas for $d(Qu)$ and $\delta_0(Qu)$ contain the same bounded terms and commutators of size $O(\rho)$ multiplying $\nabla^0 u$. They show $QV(g_0) \subset V(g_0)$. Weil invariance of the model then gives $W'u = W_0 Qu \in V(g_0) = V(g')$. □

The vanishing in (23) cannot be omitted. On the flat cone of angle 4π, use $z = w^2$ and $g_0 = |dz|^2$. A cutoff of dw is closed and coclosed near zero and belongs to $V(g_0)$. For $g_b = |dz|^2 + \mathrm{Re}(b\,dz^2)$, $0 < |b| < 1$, the metric difference is parallel, but $\delta_{g_b} dw$ has homogeneity $-3/2$ in z and is not L^2. Thus even a parallel metric perturbation can change the domain.

5.2. **Product estimates.**

Lemma 5.4. *Let B be a closed Riemannian manifold and let Y satisfy the Stokes property. Then $B \times Y$ satisfies that property and*

$$\|\nabla_B u\| \le C(\|Du\| + \|u\|). \tag{25}$$

If both factors are Kähler on their regular loci and Y has Weil invariance locally, so does the product. If $Y = C(Z)$ with admissible link, the graph-norm unit ball on a fixed compact support is precompact in $L^2(B \times C(Z))$.

Proof. Let $\mathcal{F}_\lambda$ be the finite-dimensional eigenspaces of the Hodge Laplacian on B. They are preserved by D_B and the degree sign s_B. On the product, $D = D_B \otimes 1 + s_B \otimes D_Y$. Testing weak derivatives against an eigenform on B shows that each projection u_λ lies in $\mathcal{F}_\lambda \otimes V(Y)$, and

$$\|Du_\lambda\|^2 = \lambda\|u_\lambda\|^2 + \|(1 \otimes D_Y)u_\lambda\|^2. \tag{26}$$

The cross terms cancel because $D_B s_B = -s_B D_B$ and D_Y is self-adjoint. For Stokes, take arbitrary $v \in \mathrm{Dom}\, d_{\max}$ and $w \in \mathrm{Dom}\, \delta_{\max}$, rather than assuming either lies in V. Project both onto a finite sum of the spaces $\mathcal{F}_\lambda$. The projections commute with d_B and δ_B. Their coefficient equations place the coefficients of v in $\mathrm{Dom}\, d_{Y,\max}$ and those of w in $\mathrm{Dom}\, \delta_{Y,\max}$. Stokes on Y and integration by parts on B give the adjoint identity for the projections. The projections of $v, dv, w, \delta w$ converge in L^2, so the identity holds on the full maximal domains. Summing (26) and applying the elliptic estimate on B proves (25).

If B is Kähler, W_B preserves each $\mathcal{F}_\lambda$. Use (15) for Y on a fixed compact support in (26). The resulting estimates for $(W_B \otimes W_Y)u_\lambda$ sum, proving Weil invariance. For compactness, the terms of large λ have uniformly small norm by (26); finitely many remaining terms are compact by Lemma 4.7. □

These statements apply locally when the smooth factor is not compact. For the Kähler assertion one can extend a small coordinate ball to a compact Kähler manifold: write its potential as $|z|^2 + O(|z|^3)$, cut off the cubic remainder at radius ϵ, and place it in a flat torus. The change in the Levi form is $O(\epsilon)$ and preserves positivity. For metric estimates alone a cutoff extension of the metric suffices.

5.3. **Comparison at two strata.** Let $X_0 = B \times C(Z)$ have an edge away from its vertex stratum. Assume bounded geometry at scale ρ and Weil invariance on X_0. At each edge point y, put $r_y = r(y)$. We require *uniform edge charts* with product model $P_y = B_y \times C(Z')$, where Z' is a fixed smooth closed odd-dimensional link. Their images contain $\mathbb{B}(y, \kappa r_y)$ and lie in $\mathbb{B}(y, r_y/2)$. Their domains are closed under radial contraction of the cone factor. After rescaling by r_y^{-1}, the smooth factors have uniform first-order geometry and the charts satisfy (23) with uniform constants. For the complex structures the smooth factors need only carry uniformly controlled almost complex structures. In these charts ρ is comparable to the cone radius near y.

Definition 5.5. A chart at the vertex stratum is *adapted to both strata* if it is a homeomorphism and regular-locus diffeomorphism, and its pullback tensors satisfy

$$cg_0 \le g' \le c^{-1} g_0, \quad |g' - g_0| + |I' - I_0| \le Cr, \quad |\nabla^0 g'| + |\nabla^0 I'| \le C, \tag{27}$$

and, in every specified edge chart lying in its domain,

$$g' - g_0 = t_y + e_y, \qquad I' - I_0 = t'_y + e'_y. \tag{28}$$

Here t_y, t'_y are tensors pulled back from B_y, $|t_y| + |t'_y| \le Cr_y$, $|\nabla t_y| + |\nabla t'_y| \le C$, $|t_y| < 1/2$, and $|e_y| + |e'_y| \le C\rho$.

Theorem 5.6. *Under these assumptions the chart identifies the Gaffney domains and transfers Weil invariance on a smaller neighbourhood of the vertex stratum.*

Proof. Shrink to the points x with $r(x) < r_*/2$ and $\mathbb{B}(x, 2r(x))$ contained in the chart domain, where r_* is the edge-chart radius bound. This still contains a neighbourhood of each point of the vertex stratum. Bounded geometry and (22) hold for g_0 and g' by (27).

For a form w supported in a ball of radius cr_y in an edge chart, Theorem 5.3 and (25), rescaled, give

$$r_y \|\nabla_{B_y} w\| \le C(r_y \|D_0 w\| + \|w\|). \tag{29}$$

The same bound holds with D': compare g' with the product metric $(g_{B_y} + t_y) \oplus g_{C(Z')}$. Their difference is $O(\rho/r_y)$ with derivative $O(1/r_y)$. This follows from (27), the derivative bounds on t_y, and the bounds in the model edge chart; it does not require an additional derivative assumption on e_y.

Again let $S = *_0^{-1} *'$. Its derivative is bounded and $|S - 1| \le Cr$. Where $\rho \ge cr$, the first-order sum in (24) is controlled by $C\rho|\nabla^0 u|$. In an edge chart, smooth dependence of the Hodge star on its metric and (28) give

$$S = S_{B_y} \otimes 1 + O(\rho), \qquad |S_{B_y} - 1| \le Cr_y. \tag{30}$$

To check the remainder, compare the pair of metrics with $(g_{P_y}, g_{P_y}+t_y)$. The error is bounded by $|e_y| + |g_0 - g_{P_y}||t_y| \le C\rho$; the second product is $O((\rho/r_y)r_y)$. Contraction in a normal direction commutes with $S_{B_y} \otimes 1$. The first-order sum is consequently bounded pointwise by

$$C\big(r_y|\nabla_{B_y} u| + \rho|\nabla^0 u| + |u|\big). \tag{31}$$

Changing connections costs $O(1/r_y)$, multiplied by $O(r_y)$, which explains the last term.

Cover $\{\rho < cr\}$ by edge balls of radii comparable to r_y with bounded overlap, using doubling. Their cutoffs have derivative $O(1/r_y)$. Apply (29) to the cutoffs of u and sum their squared estimates; cutoff errors are bounded by $C\|u\|^2$. Equation (22) controls the normal term in (31). Thus the right side of (24) is L^2, bounded by the graph norm of either D_0 or D'. This proves both domain inclusions. The choice of smaller neighbourhood ensures that the charts used in this cover lie inside the original chart domain: a nearest singular point to x with $\rho(x) < cr(x)$ is an edge point y with $r_y \asymp r(x)$.

Finally $Q = W_0^{-1}W'$ has the same properties as S and the same decomposition (30). In a product chart the Weil operator is a tensor product, and its dependence on I is polynomial. Applying the preceding estimates to its commutators with wedge product and contraction gives $QV(g_0) \subset V(g_0)$. The last step of Theorem 5.3 transfers Weil invariance. □

6. Quotient charts and the singular link

We construct the charts required by Section 5. The construction first treats a stabiliser acting freely on the punctured zero fibre. It applies both along $\Sigma \setminus \Delta$ and along the edge of C_2.

6.1. **A finite-dimensional slice with its metric.** Let $\mathcal{X}$ be a flat hyperkähler affine space and let $\mathcal{G}$ act preserving the structure, with moment map μ. We allow either a finite-dimensional space and compact group, or the space of unitary connections on our K3 surface. In the latter case choose Sobolev order $s \ge 3$ for connections and $s+1$ for gauge transformations, with the L^2 metric, and take $\mu(A) = (F_A^0)^+$. Fix $x_* \in \mu^{-1}(0)$, its compact stabiliser H, and an H-invariant complement $\mathfrak{q}$ of $\mathfrak{h}$ in the gauge Lie algebra. The four spaces $\mathfrak{q}x_*, I_1\mathfrak{q}x_*, I_2\mathfrak{q}x_*, I_3\mathfrak{q}x_*$ are mutually orthogonal: the off-diagonal pairings are components of the moment map evaluated on brackets, and vanish. Their orthogonal complement is a quaternionic representation $\mathbb{E}$. Write $\mathbb{E} = \mathbb{E}_0 \oplus \mathbb{E}_1$, where $\mathbb{E}_0 = \mathbb{E}^H$.

Proposition 6.1. *There is a real-analytic equivariant map*

$$\iota(y) = x_* + y + c(y), \qquad c(y) \in \sum_a I_a\mathfrak{q}x_*, \quad c(y) = O(|y|^2),$$

with $\mu^{\mathfrak{q}}\iota = 0$. Put $\widehat{\mu} = \mu^{\mathfrak{h}}\iota$ and $Y = \widehat{\mu}^{-1}(0)$. Then Y/H is a neighbourhood of $[x_*]$ in the quotient, preserving stabilisers. There are smooth equivariant quaternionic Hermitian tensors $(\widehat{g}, \widehat{I}_a)$ on $\mathbb{E}$, equal to the flat tensors at zero, with these properties:

1. *on Y, for every $v \in T_y\mathbb{E}$ and $\xi \in \mathfrak{h}$,*

$$\langle d\widehat{\mu}_a(v), \xi\rangle = \widehat{g}(\widehat{I}_a\xi y, v); \tag{32}$$

2. *along $\mathbb{E}_0$, $\widehat{\mu} = d\widehat{\mu} = 0$, and its quadratic term in $\mathbb{E}_1$ is the flat moment map for the fibre structure $(\widehat{g}_u, \widehat{I}_a(u))|_{\mathbb{E}_1}$;*
3. *the tensors restricted to $\mathbb{E}_0$ are hyperkähler, and their quotient tensors on the free part of Y/H are the original quotient metric and complex structures.*

If $H' \subset H$ and $\mathfrak{h}^{H'} \subset \mathfrak{h}'$, then $\widehat{\mu}$ vanishes on $\mathbb{E}^{H'}$.

Proof. The derivative of $\mu^{\mathfrak{q}}$ in $\sum I_a\mathfrak{q}x_*$ is invertible. In finite dimensions the implicit function theorem and the ordinary slice theorem give c and the local quotient. For connections, write $d = d_{x_*}$. The K3 identities and $(F^0_{x_*})^+ = 0$ give, with positive normalisation constants,

$$(d^+)^*(\phi_a) = c_+ \sum_a I_a d\phi_a, \qquad d^+(d^+)^* = c'_+(d^*d)\otimes 1. \tag{33}$$

For the off-diagonal components one uses $d^+d\phi = [F^+_{x_*}, \phi] = 0$ in the trace-free complex. Thus $\mathbb{E} = \ker d^* \cap \ker d^+$ is finite dimensional, the cokernel is $\mathfrak{h}\otimes\mathbb{R}^3$, and d^*d is invertible on $\mathfrak{q}$. Writing $c = (d^+)^*\phi$, the equation is

$$\Pi_{\mathfrak{q}}\{d^+(d^+)^*\phi + ((y + (d^+)^*\phi)\wedge(y + (d^+)^*\phi))^+\} = 0.$$

It is analytic from order $s+1$ to order $s-1$, with invertible derivative in ϕ. The implicit function theorem and elliptic regularity give the claimed finite-dimensional analytic map, with values in every Sobolev order after shrinking its domain. The Coulomb slice theorem [DK90, Proposition 4.2.9] identifies the quotient and its stabilisers.

At $x = \iota(y)$ let p_y project orthogonally off the quaternionic span of $\mathfrak{q}x$. This span is invariant under the I_a even when its four summands are not orthogonal. In gauge theory its projection is analytic: the map

$$Q_y(\phi_0, \phi_a) = d_x\phi_0 + \sum_a I_a d_x\phi_a$$

on $\mathfrak{q}\otimes\mathbb{H}$ is uniformly bounded below near $y = 0$, and $p_y = 1 - Q_y(Q_y^*Q_y)^{-1}Q_y^*$. The inverse follows from the coercive form and elliptic regularity. Its complement has dimension $\dim\mathbb{E}$. Therefore $d^\perp\iota = p_y d\iota$ is invertible onto that complement, and we define

$$\widehat{g}(v,w) = g_{\mathcal{X}}(d^\perp\iota v, d^\perp\iota w), \qquad \widehat{I}_a = (d^\perp\iota)^{-1} I_a d^\perp\iota.$$

On Y the four orbit-direction summands are orthogonal. Since $\mu^{\mathfrak{q}}\iota = 0$, the difference $d\iota v - d^\perp\iota v$ is in $\mathfrak{q}x$ there. It is orthogonal to $I_a\mathfrak{h}x$. Equivariance and the ambient moment identity give (32). On $\mathcal{X}^{H'}$ the components of μ along $\mathfrak{h}'$ are constant and zero. Equivariance puts their remaining values in the dual of $\mathfrak{h}^{H'}$; the stated condition on H' therefore forces all components to vanish. This proves the last assertion and gives $\widehat{\mu} = 0$ on $\mathbb{E}_0$.

At $u\in\mathbb{E}_0$ the linear term in the $\mathfrak{h}$ component vanishes, because $\xi\iota(u) = 0$ for $\xi\in\mathfrak{h}$. Its quadratic term is the ambient flat moment map on $d\iota_u\mathbb{E}_1$. Replacing $d\iota_u n$ by $d^\perp\iota_u n$ does not change it. The mixed term vanishes by $[\mathfrak{h},\mathfrak{q}]\subset\mathfrak{q}$ and orthogonality to $I_a\mathfrak{q}x$; the pure orbit term is a component of $\mu(x)$ evaluated on a double bracket. This proves (2).

For the metric, minimise first over $\mathfrak{q}x$ and then over $\mathfrak{h}x$. The result is $\min_{\xi\in\mathfrak{h}}\widehat{g}(v-\xi y, v-\xi y)$. The corresponding horizontal lift is $d^\perp\iota(Pv)$, so the same calculation identifies the complex structures. Along the fixed locus, the stratum is locally the smooth hyperkähler quotient of $\mathcal{X}^H$ by the centraliser of H modulo its ineffective part. This proves (3). □

6.2. **Normalising the fibres.** We next work abstractly with the tensors and map in Proposition 6.1. Suppose H is free on the connected punctured zero fibre of the flat moment map μ_0 of $\mathbb{E}_1$. The fixed coordinates are $u\in\mathbb{E}_0$, the normal coordinates $z\in\mathbb{E}_1$, and $\rho = |z|$.

Lemma 6.2. *There is an equivariant $C^{1,1}$ coordinate change Ξ, smooth off $\mathbb{E}_0$ and equal to the identity there, carrying $Y_0 = \mathbb{E}_0\times\mu_0^{-1}(0)$ onto Y. The pulled-back tensors are Lipschitz, agree exactly with the flat tensors in all normal blocks on $\mathbb{E}_0$, and induce the quotient structure of Y/H.*

Proof. First identify the varying quaternionic Hermitian structures on the normal fibres with a fixed one. The intertwiner $B_u = \frac{1}{4}(1 - \sum_a I_a \widehat{I}_a(u))$ is invertible near zero and commutes with H. Pulling the metric through B_u gives a positive operator G_u commuting with the I_a and H. Thus $A_u = G_u^{1/2} B_u$ is an equivariant quaternionic isometry from the varying fibre to the flat fibre. In coordinates $(u, A_u^{-1} z)$ the map becomes

$$\mu_1(u, z) = \mu_0(z) + R(u, z), \qquad R = O(|z|^3).$$

Near the punctured flat zero fibre, $d\mu_0(z)$ has an equivariant right inverse S_z of homogeneity -1. Solve

$$b = S_z\{R(u, z) - \mu_0(b)\}.$$

Contraction gives $b = O(|z|^2)$ and $\mu_0(z + b) = \mu_1(u, z)$. After radial rescaling, differentiation of this equation gives $|\partial^i b| \le C_i |z|^{2-i}$. An angular invariant cutoff extends the correction off a conic neighbourhood of the zero fibre. The estimate $|\mu_0(z)| \ge c|z|^2 \operatorname{dist}(z/|z|, \mu_0^{-1}(0))$ near that fibre shows that every sufficiently small zero of μ_1 lies where the cutoff is one. Hence the inverse of $(u, z) \mapsto (u, z + b)$, followed by the fibre isometry, gives Ξ. The second derivative bound gives $C^{1,1}$ across $\mathbb{E}_0$. The differential there is $\mathrm{id} \oplus A_u^{-1}$, proving exact agreement of the normal blocks. Equation (32) ensures that the pulled-back complex structures preserve the horizontal tangent space on Y_0. □

Lemma 6.3. *Suppose invariant Lipschitz tensors $(\widehat{g}, \widehat{I}_a)$ on Y_0 arise as above, agree with the flat tensors in normal blocks along $\mathbb{E}_0$, and preserve the horizontal tangent spaces. Their quotients differ by $O(\rho)$ from the product of their restrictions to $\mathbb{E}_0$ with the flat cone quotient, with bounded covariant first derivative. If two such sets of tensors differ by $O(t)$ in Lipschitz norm and share the normal blocks, then their quotient difference, after subtracting the difference on the smooth factor, is $O(t\rho)$ with derivative $O(t)$.*

Proof. Let Π_0 be the flat projection to $\mathfrak{h}z$, $P_0 = 1 - \Pi_0$, and write $\widehat{g} = g_{\mathrm{fl}} + \tau$. On a conic neighbourhood of the zero fibre, $|\partial \Pi_0| \le C/\rho$. Its image and derivatives are normal to $\mathbb{E}_0$, and it kills fixed directions. Normalisation gives

$$|\tau(v_1, \cdot)| \le C\rho |v_1|, \quad |(\widehat{I}_a - I_a)v_1| \le C\rho |v_1|, \quad |((\widehat{I}_a - I_a)v)_1| \le C\rho |v|.$$

The metric-orthogonal orbit projection satisfies

$$\Pi - \Pi_0 = \big(\Pi_0(1 + \tau^\sharp)|_{\mathfrak{h}z}\big)^{-1} \Pi_0 \tau^\sharp P_0.$$

It follows that $|\Pi - \Pi_0| \le C\rho$ and $|\partial(\Pi - \Pi_0)| \le C$. Every derivative of Π_0 is multiplied by a normal block of size $O(\rho)$.

The basic lift of the quotient metric is $\widehat{g}(P\cdot, P\cdot)$, where $P = 1 - \Pi$. Subtracting the flat lift leaves its fixed-factor restriction plus $O(\rho)$; differentiating gives a bounded tensor by the projection estimates. Passing from ambient derivatives to derivatives on Y_0 introduces its second fundamental form, of size $O(1/\rho)$. Its values are normal, so it too multiplies an $O(\rho)$ block. Riemannian submersion then bounds the covariant derivative on the quotient. For complex structures use the lift $P_0 \widehat{I}_a P$ and the same estimates.

For the relative assertion interpolate the two metrics by $g_s = g_* + s\tau_*$. If P_s is its horizontal projection, stationarity of the minimum over orbit directions gives

$$\frac{d}{ds} g_s(P_s v, P_s w) = \tau_*(P_s v, P_s w).$$

The normal blocks of τ_* vanish on the fixed locus and are $O(t\rho)$; the preceding derivative estimates now all carry a factor t. For the complex structures use

$$\frac{d}{ds}\Pi_s = \Pi_s g_s^{-1}\tau_* P_s$$

and split their lift difference into the change of complex structure and the change of projection. Each has the same factor t and normal vanishing. This proves both estimates. □

Together, the two lemmas construct adapted charts for each complex structure, with model the actual stratum metric times the flat transverse cone. The quotient map is a homeomorphism for quotient topologies, smooth on regular loci. Its metric comparison also identifies completions. The smooth factor may be completed to a compact Kähler manifold as in Section 5.

6.3. **The link Z_2 and induction on depth.** Apply these charts first in finite dimensions to the rank-one stratum of C_2, using (13), and then in gauge theory to $\Sigma \setminus \Delta$. In both cases the normal circle action is free off zero and the local model is $B \times C_1$.

Proposition 6.4. *The link Z_2 is admissible. The Stokes property holds on M, and the Gaffney domain is Weil invariant on C_2 and on $M \setminus \Delta$, for all three complex structures. The models $B \times C_2$ have bounded geometry at scale ρ and uniform edge charts.*

Proof. Stokes is invariant under quasi-isometry, since the minimal and maximal graph norms of d are equivalent. It is local for forms supported away from the artificial boundary of a chart, and on a compact space these local statements combine by Lipschitz partitions of unity. The odd smooth link Z_1 gives Stokes on C_1; the product lemma gives it on $B \times C_1$. The adapted charts therefore give Stokes on $C_2 \setminus \{0\}$ locally and on $M \setminus \Delta$ locally. Annular cone coordinates are quasi-isometric to interval products with Z_2. Applying local Stokes on a circle product and then to forms pulled back from Z_2 proves Stokes on Z_2.

To prove compact resolvent, take a sequence bounded in $V(Z_2)$ and pull it back to the cone, multiplying by a fixed radial cutoff in $1 < r < 2$. The cone differential formulas give a bounded sequence in $V(C_2)$ with fixed compact support. A finite cover by adapted $B \times C_1$ charts transfers its graph norm to product graph norms. Lemma 5.4 makes each local piece precompact. Its cone L^2 norm is a fixed positive multiple of its link norm, degree by degree. Thus $V(Z_2) \hookrightarrow L^2(Z_2)$ is compact. Oddness of $\dim Z_2$ now gives Stokes on C_2, proving admissibility. The imported quasi-isometry (10) transfers Stokes from $B \times C_2$ to M near Δ; compactness of M completes the global assertion.

For Weil invariance, use the smooth-link cone theorem and the product lemma on $B \times C_1$, then Theorem 5.3 in the adapted charts. This treats $C_2 \setminus \{0\}$ and $M \setminus \Delta$. Theorem 4.5 applies to the now admissible Z_2 and treats the vertex of C_2.

Finally the distance ρ on C_2 is homogeneous. On the unit annulus, finitely many adapted edge charts and smooth charts supply bounded geometry; dilations give it at all scales. For doubling, balls of radius comparable to their distance from the vertex have volume comparable to the appropriate power of their radius, by homogeneity and compactness of the link. Smaller balls reduce to the unit annulus and the doubling product charts. Edge charts on that annulus dilate to uniform edge charts at every radius. Taking products with small balls in B, extended to compact smooth factors, proves the assertion for $B \times C_2$. □

7. A chart at the diagonal

The construction of Section 6 cannot be applied directly at Δ: the SO(3) representation has rank-one zeros with circle stabiliser. We construct a chart near that rank-one locus and a second chart away from it, then join them through the free part. The normalisation near rank one is what ensures the stronger normal estimate in (28).

7.1. **Scaling the slice.** Use Proposition 6.1 at $p \in \Delta$ and perform only the fibrewise linear normalisation of Lemma 6.2, which requires no freeness. In

$$\mathbb{E} = \mathbb{E}_0 \oplus (\mathbb{R}^3 \otimes \mathbb{H}^{k+1}), \qquad y = (u, n), \qquad r = |n|,$$

the resulting data $\mathcal{D} = (\widehat{g}, \widehat{I}_a, \widehat{\mu})$ have flat normal blocks along $\mathbb{E}_0$, satisfy (32) on their zero set, and obey

$$\widehat{\mu} = \mu_0 + O(r^3), \qquad \widehat{\mu}|_{\mathbb{E}^T} = 0$$

for every maximal circle $T \subset \mathrm{SO}(3)$. The latter follows from $\mathfrak{so}(3)^T = \mathfrak{t}$ in Proposition 6.1. Compare with the product data

$$\mathcal{D}_* = (g_B \oplus g_{\mathfrak{fl}}, I_{a,B} \oplus I_a, \mu_0),$$

where $(g_B, I_{a,B})$ is the restriction to $\mathbb{E}_0$. Their quotient is $X_0 = (\mathbb{E}_0, g_B) \times C_2$.

At a normal radius $\bar{r} > 0$, use coordinates $(u - u_0, n)/\bar{r}$ and divide the moment map by $\bar{r}^2$. Smoothness, normal agreement, and the cubic remainder imply

$$\|\mathcal{D} - \mathcal{D}_*\|_{C^i} \le C_i \bar{r} \tag{34}$$

on fixed-size scaled sets, for every fixed i. This includes the moment map and its derivatives: every term in its remainder has at least three normal factors.

7.2. **Separating the circle stabiliser.** Fix a unit vector $e \in \mathbb{R}^3$, its stabiliser T, and $N(T) \cong O(2)$. Write $\mathfrak{so}(3) = \mathbb{R}e \oplus e^\perp = \mathfrak{t} \oplus \mathfrak{p}$. For $n : \mathbb{R}^3 \to \mathbb{H}^{k+1}$ let $s_1 \ge s_2 \ge s_3$ be its singular values and put $\rho_T = (s_2^2 + s_3^2)^{1/2}$. This is the Euclidean distance to rank at most one. For a fixed $c_0 < 1$, the region $\rho_T < c_0 s_1$ is

$$\mathrm{SO}(3) \times_{N(T)} \{(u, l, n'') : l \neq 0,\ n'' \in e^\perp \otimes l^{\perp_{\mathbb{R}}},\ |n''| < c_0 |l|\}. \tag{35}$$

Indeed the largest eigenline of the Gram matrix is unique there. Move it to $\mathbb{R}e$; the remaining ambiguity is precisely $N(T)$. In these coordinates $n = e \otimes l + n''$, $s_1 = |l|$ and $\rho_T = |n''|$.

Decompose the last normal space as

$$e^\perp \otimes l^{\perp_{\mathbb{R}}} = F_l \oplus (e^\perp \otimes \mathrm{Im}(\mathbb{H})l), \qquad F_l = e^\perp \otimes (\mathbb{H}l)^\perp.$$

The group T acts on F_l as on $T^*\mathbb{C}^{2k}$. Let $\mathfrak{B} = \mathbb{E}_0 \times (\mathbb{H}^{k+1} \setminus \{0\})$ and let $\mathfrak{F}$ be the bundle of these F_l with $|f| < c_0|l|$. If $f = e_2 \otimes f_2 + e_3 \otimes f_3$, the flat equations along $\mathfrak{p}$ impose orthogonality to $I_a l$, and the remaining equations are

$$\mu_T(f)_a = g_{\mathfrak{fl}}(I_a f_2, f_3) = 0.$$

Consequently, with $Y_0 = \mathbb{E}_0 \times \mu_0^{-1}(0)$,

$$(Y_0 \cap \{\rho_T < c_0 s_1\})/\mathrm{SO}(3) = \{(b, f) \in \mathfrak{F} : \mu_T(f) = 0\}/N(T). \tag{36}$$

On this part of X_0, the distance to the singular set equals ρ_T. The Euclidean distance is a lower bound for the quotient length, while the path $(u, e \otimes l + tf)$ lies in Y_0 and attains that bound.

Lemma 7.1. *For $c_0 < 1/\sqrt{2}$ and r sufficiently small, the equations along $\mathfrak{p}$ determine a smooth equivariant correction $c^\flat(b,f) \in e^\perp \otimes \mathrm{Im}(\mathbb{H})l$. The map $j(b,f) = (u, e\otimes l + f + c^\flat(b,f))$ carries the zero set of $\mu^\flat = (\widehat{\mu}\circ j)^{\mathfrak{t}}$ to Y in the slice (35). Projection off the quaternionic span of $\mathfrak{p}j(b,f)$ induces smooth quaternionic Hermitian tensors $(g^\flat, I_a^\flat)$ on $\mathfrak{F}$. On $\{\mu^\flat = 0\}$ they satisfy the moment identity for T, and their circle quotient is the original $\mathrm{SO}(3)$ quotient. Along $\mathfrak{B}$ their moment map has the quadratic term given by their fibre structure.*

Proof. The derivative in $c^\flat$ of the flat $\mathfrak{p}$ equations is the map with components $g(I_a l, c_j^\flat)$, $j = 2,3$. It is invertible with inverse of size $C/|l|$. Equation (34) and the implicit function theorem give $c^\flat$, with scaled C^i norm $O(\bar{r})$. It vanishes at $f = 0$ since $\widehat{\mu}$ does on $\mathbb{E}^T$. Every nearby zero in the singular-value slice has this form, by the same uniqueness. For the product data the correction is identically zero.

Let $Q_z = \mathfrak{p}z + \sum_a \widehat{I}_a \mathfrak{p}z$ and let p_z project off it. We check that $p_z dj$ is invertible. In the flat model its possible kernel consists of tangent vectors of the form

$$\dot{l} = \alpha_2 f_3 - \alpha_3 f_2, \qquad \dot{f}_2 = \alpha_3 l, \qquad \dot{f}_3 = -\alpha_2 l \quad (\alpha_2, \alpha_3 \in \mathbb{H}).$$

Differentiating $f_j \perp \mathbb{H}l$ gives

$$(|\alpha_2|^2 + |\alpha_3|^2)|l|^2 \le 2|f|^2(|\alpha_2|^2 + |\alpha_3|^2).$$

Thus the kernel is zero for the stated c_0. Dimensions agree: Q_z has real dimension eight. Homogeneity and compactness give a uniform inverse at scale $|l|$, and (34) preserves invertibility for the actual data. Define

$$g^\flat(v,w) = \widehat{g}(p_z dj v, p_z dj w), \qquad I_a^\flat = (p_z dj)^{-1}\widehat{I}_a p_z dj.$$

They differ from the corresponding model data by $O(\bar{r})$ in scaled C^i norm.

At a zero z, the four quaternionic orbit subspaces are orthogonal. Since the $\mathfrak{p}$ moment equations vanish identically on the image of j, the component of djv in Q_z lies in $\mathfrak{p}z$. It is orthogonal to $\widehat{I}_a \mathfrak{t}z$. Equivariance therefore gives

$$\langle d\mu_a^\flat(v), \xi\rangle = g^\flat(I_a^\flat \xi(b,f), v) \quad (\xi \in \mathfrak{t}).$$

Minimising the metric first over $\mathfrak{p}$ and then over $\mathfrak{t}$ identifies the quotient metric, and the same horizontal lift identifies the complex structures. No reduction by a subgroup with Lie algebra $\mathfrak{p}$ is asserted.

It remains to verify the Hessian along $f = 0$. There $\mu^\flat = d\mu^\flat = 0$. Let Q_b be its quadratic term. It is $O(\bar{r})$-close to μ_T after scaling. Every regular nonzero zero f of Q_b continues to a curve $(b, tf(t))$ of zeros of $\mu^\flat$ with $f(0) = f$, by dividing the equation by t^2 and applying the implicit function theorem. Differentiating the preceding moment identity along this curve gives

$$\partial^2 \mu_a^\flat(b)(f,f') = g_b^\flat(I_a^\flat \xi f, f').$$

Choose a finite spanning collection of regular zeros of the flat circle moment map. Such a collection exists: in $T^*\mathbb{C}^{2k}$ the vectors $(\alpha z_i, \beta z_j^*)$ with $i \ne j$ and $|\alpha| = |\beta| = 1$ are zeros and span the real vector space. They persist and still span for Q_b. The displayed identity, linear in f, therefore holds for every f, proving the quadratic-term assertion. □

7.3. **The chart near rank one.** Over a ball of radius $c|l_0|$ in $\mathfrak{B}$, identify F_{l_0} with F_l by the polar part of orthogonal projection $(\mathbb{H}l_0)^\perp \to (\mathbb{H}l)^\perp$, tensored with $e^\perp$. Denote this quaternionic trivialisation by ϑ. It commutes with T and has uniform scaled derivatives. The fibrewise construction of Lemma 6.2, now with parameter $b = (u,l)$, gives a map $\Xi^\flat$ which

fixes $\mathfrak{B}$, takes $\mu_T^{-1}(0)$ to $(\mu^\flat)^{-1}(0)$, and normalises the fibre metric and all three complex structures there. Its deviation from the identity has scaled $C^{1,1}$ norm $O(\bar r)$.

Here is why that estimate is uniform even at $f = 0$. After the linear fibre isometry, the residual map is $\mu_T + R^\flat$, with

$$|\partial^i R^\flat| \le C_i \bar r |f|^{3-i}$$

in scaled coordinates. The right inverse of $d\mu_T(f)$ has size $C/|f|$. The same contraction equation as in Lemma 6.2 gives a correction with derivatives $C_i \bar r |f|^{2-i}$. Its second derivative is uniformly bounded; away from $f = 0$ all orders are controlled. Every operation is canonical for the invariant data, hence equivariant under $N(T)$.

For the model data this map is the identity. Both its reduced tensors and the pulled-back actual reduced tensors have the same normal blocks along $\mathfrak{B}$. Their difference is $O(\bar r)$ in scaled Lipschitz norm. The relative part of Lemma 6.3 gives

$$(\Theta^{\mathrm{in}})^* g - g_0 = t_b + O(\rho_T), \qquad |t_b| \le C\bar r, \quad |\nabla t_b| \le C, \quad |\nabla^0((\Theta^{\mathrm{in}})^* g - g_0)| \le C, \tag{37}$$

and the same assertions for each complex structure. Here Θ^{in} is induced by $j\Xi^\flat$, and t_b is pulled back from the smooth rank-one stratum. To check the scaling, the relative lemma gives an error $C\bar r\widehat\rho$ at unit scale. Since $\widehat\rho = \rho_T/\bar r$, it becomes $C\rho_T$. A covariant derivative costs $\bar r^{-1}$, cancelling the factor $\bar r$.

Applying the absolute part of the same lemma to the model reduced tensors gives uniform edge charts $B_b \times C_1 \to X_0$. The smooth factors may be extended to compact factors with uniform scaled geometry by cutoffs. The complex structure is extended as an almost complex structure, which is all the edge comparison uses. For example, write it locally as $AI(b_0)A^{-1}$ with A close to the identity and replace A by $1 + \chi(A - 1)$. The tensors t_b are extended with the same cutoff. Thus (37) is exactly the normal and tangential decomposition needed in Definition 5.5.

7.4. **Joining the charts.** Choose $c_3 > 0$ smaller than a quarter of the radius of validity of the inner construction. On the cone $\rho_T > c_3 s_1/2$ the action is free near the flat zero set. The construction of Lemma 6.2 gives a map Ξ_2 with $\mu_0\Xi_2 = \widehat\mu$ and scaled C^i distance $O(\bar r)$ from the identity. Its inverse Ξ^{out} carries Y_0 to Y in this region.

Extend the inner map to ambient slice coordinates by

$$(b, f, c') \longmapsto (u, e \otimes l + f' + c^\flat(b, f') + c'), \qquad (b, f') = \Xi^\flat(b, f),$$

and then equivariantly using (35). It is smooth on the regular zero fibre, fixes the rank-one locus, and has scaled C^i distance $O(\bar r)$ from the identity on the free overlap. No smoothness of this ambient extension is needed at points with $f = 0$ and $c' \ne 0$. On $c_3 s_1 \le \rho_T \le 2c_3 s_1$, set $\Gamma = \Xi_2\Xi^{\mathrm{in}}$. This map preserves Y_0.

Choose a cutoff φ of ρ_T/s_1 equal to one at the inner end and zero at the outer end. The naive interpolation $x + \varphi(\Gamma x - x)$ need not preserve the zero set. Correct it to

$$G(x) = x + \varphi(\Gamma x - x) + b(x)$$

by requiring

$$\mu_0(G(x)) = (1 - \varphi)\mu_0(x) + \varphi\mu_0(\Gamma x). \tag{38}$$

Take $b(x)$ in the image of the right inverse of $d\mu_0(x)$. Before correction the error in (38) is $(\varphi^2 - \varphi)\mu_0(\Gamma x - x)$, because μ_0 is quadratic. It is $O(\bar r^2)$ in scaled C^i norm. The right inverse is uniformly bounded at unit scale on this free overlap. Contraction therefore gives a unique correction with that bound. It vanishes where φ is zero or one. Consequently the map

$$\Xi = \Xi^{\mathrm{in}} \quad (\rho_T \le c_3 s_1), \qquad \Xi = \Xi^{\mathrm{out}} G \quad (c_3 s_1 \le \rho_T \le 2c_3 s_1), \qquad \Xi = \Xi^{\mathrm{out}} \quad (\rho_T \ge 2c_3 s_1)$$

is well defined and carries Y_0 into Y.

We verify that it is a quotient chart. In the outer region and overlap its unscaled estimates are

$$|\Xi - \mathrm{id}| \le Cr^2, \qquad |d\Xi - 1| \le Cr, \qquad |d^2\Xi| \le C.$$

On each rescaled free annulus it is a local diffeomorphism close to the identity. Uniform local injectivity and its small displacement imply injectivity on that region. In the inner region injectivity follows from the unique singular-value slice and the fibrewise coordinate changes. The smaller inner region $\rho_T/s_1 < c_3/2$ and the region $\rho_T/s_1 > c_3$ have images separated by $c_3/2 - O(r)$ in this ratio; the remaining overlap is already covered by the free-annulus argument. The equivariant map thus induces an injective quotient map. It is open away from the vertex stratum, by the local diffeomorphism property on the free part and the explicit inner product charts near rank one.

It is also locally onto. On a small set $|u| < t_1$, $0 < r < t_2$, the estimate $|\Xi - \mathrm{id}| \le Cr^2$ and preservation of u make the map proper over $|u| < t_1$, $0 < r < t_2/2$. Its image there is open and closed. The target is locally a normal analytic space; a small connected neighbourhood with its proper analytic diagonal removed is connected. It meets the image, so is contained in it. At $r = 0$ both the map and its inverse extend continuously by the same quadratic displacement estimate. Stabilisers are preserved: rank-one points are fixed, and the nonzero circle-normal coordinate stays nonzero. This proves the homeomorphism and regular-locus diffeomorphism assertions.

Theorem 7.2. *At every point of Δ, M admits a chart adapted to both strata, for all three complex structures, with model $(\mathbb{E}_0, g_B) \times C_2$.*

Proof. Compose the quotient map just constructed with the initial fibre normalisation and the gauge slice map. In the outer region Lemma 6.3 gives $O(r)$ error and bounded first derivative. In the inner region (37) gives these bounds and (28). An edge chart may also meet the outer region; there $\rho \ge c_3 s_1 \asymp r$, so the same smooth tensor t_b leaves an error $O(r) = O(\rho)$. Thus the decomposition holds throughout each edge chart. Shrinking r makes the metrics uniformly equivalent and $|t_b| < 1/2$. The compact extension of the smooth vertex factor is available for each Kähler structure. All conditions of Definition 5.5 follow. □

Corollary 7.3. *The Gaffney domain of (M_{reg}, g) is Weil invariant for each I_a. Its harmonic forms carry the pointwise $\mathfrak{so}(4,1)$ action.*

Proof. Proposition 6.4 gives Weil invariance away from Δ and on the models $B \times C_2$. Theorem 5.6 transfers it through Theorem 7.2. Locality covers M. The Stokes property was proved in Proposition 6.4; apply Proposition 4.2 to the three structures and use (4.1). □

8. Harmonic representatives and intersection cohomology

8.1. **The local de Rham complex.** Let $\mathcal{L}^j$ be the sheaf of forms u on M_{reg} for which u and its weak differential are square integrable on the regular part of every compact subset of the open set in question. Lipschitz partitions of unity make these sheaves fine. Since M is compact, their global complex is $\operatorname{Dom} d_{\max}$. Quasi-isometric regular-locus diffeomorphisms identify these complexes of sheaves.

We recall explicitly the two local calculations needed. For an interval factor t, write $u = \alpha + dt \wedge \beta$ and choose $\phi \in C_c^\infty$ with integral one. The maps

$$Au = \int \phi(t_0)\alpha(t_0)\, dt_0, \qquad Ku(t) = \int \phi(t_0) \int_{t_0}^{t} \beta(s)\, ds\, dt_0$$

satisfy $dK + Kd = 1 - \mathrm{pr}^* A$ and $dA = Ad$. They are bounded on compact subintervals, and globally on bounded intervals. Testing against smooth forms proves the identities for weak derivatives. Iteration removes a cube factor. For a truncated cone with odd-dimensional link Z and finite-dimensional L^2 cohomology, the cone formula is

$$H^j_{(2)}(C_R(Z)) = \begin{cases} H^j_{(2)}(Z), & j < (f+1)/2, \\ 0, & j \geq (f+1)/2, \end{cases} \qquad f = \dim_{\mathbb{R}} Z. \tag{39}$$

In the first line the map is pullback. This is Cheeger's formula in the generality of [BHS92, Proposition 2.4], applied to Z_{reg}; regularisation identifies the smooth and weak de Rham complexes used in the two formulations. In particular, the equality degree belongs to the vanishing range.

At a point of the vertex stratum of a product cone, the local cohomology sheaf of $\mathcal{L}^\bullet$ is therefore $H^j_{(2)}(Z)$ up to $(f-1)/2$ and zero above it. These identifications are locally constant along the stratum. On deleting that stratum the local complex is an interval product with Z, so its derived direct image has stalk $H^j_{(2)}(Z)$ in every degree. The attaching map is the identity in the lower range, because all the identifications are induced by the same pullback. One may compute these stalks either from locally L^2 sections or from globally L^2 sections on smaller truncated neighbourhoods: the restriction maps between the two systems interleave, hence have the same direct limit.

Theorem 8.1. *There is a canonical isomorphism*

$$\Phi : IH^\bullet(M;\mathbb{R}) \longrightarrow H^\bullet_{(2)}(M_{\mathrm{reg}}, g).$$

The maximal exterior derivative has closed range, and every cohomology class has a unique harmonic representative. If $\mathcal{A}^\bullet$ is the sheaf of locally bounded forms with locally bounded weak differential, then it resolves $\mathbb{R}_M$, and the product $\mathcal{A}^\bullet \otimes \mathcal{L}^\bullet \to \mathcal{L}^\bullet$ realises the ordinary-cohomology action under Φ.

Proof. On the regular locus, $\mathcal{L}^\bullet$ resolves $\mathbb{R}$. Along $\Sigma \setminus \Delta$ use the imported chart with link Z_1, and along Δ the chart with link Z_2. Proposition 6.4 supplies finite-dimensional cohomology of the latter. The real codimensions are $8k-4$ and $12k$, so the middle perversities are $4k-3$ and $6k-1$. These are exactly $(f-1)/2$ for the respective links. The preceding local calculation verifies the constructibility, stalk vanishing, and attachment axioms of the intersection complex [GM83, §3.3, Theorem 3.5]. With the convention that its restriction to the regular locus is in degree zero,

$$\mathcal{L}^\bullet \simeq \mathcal{IC}_M.$$

The identification restricts to the identity on the constant sheaf there and is canonical. Fineness gives Φ.

Finite-dimensional intersection cohomology implies closed ranges in this Hilbert complex. Here is the functional-analytic point. For a closed differential d whose algebraic range has finite codimension in the closed space $\ker d_{\mathrm{next}}$, choose a finite-dimensional algebraic complement F in that kernel. The bounded map $\mathrm{Dom}\, d \oplus F \to \ker d_{\mathrm{next}}$, $(u, f) \mapsto du + f$, from the graph-norm Banach space is surjective. The open mapping theorem makes the projection to F continuous; its kernel is $\mathrm{ran}\, d$, which is therefore closed. Stokes and Hilbert complex theory then give the orthogonal decomposition $L^2 = \mathcal{H} \oplus \mathrm{ran}\, d \oplus \mathrm{ran}\, \delta$ and uniqueness of harmonic representatives.

For $\mathcal{A}^\bullet$, normality makes sufficiently small regular loci connected, so the degree-zero kernel is $\mathbb{R}$. In a product-cone chart, simultaneous radial contraction m_t in all factors gives, for a

bounded closed j-form with $j > 0$,

$$K\alpha = \int_0^1 m_t^*(\iota_{r\partial_r}\alpha)\,\frac{dt}{t}, \qquad |K\alpha| \le \frac{r}{j}\|\alpha\|_\infty.$$

The distributional homotopy identity gives $dK\alpha = \alpha$; $m_t^*\alpha \to 0$ since $j > 0$. Quasi-isometry transfers this to the actual chart. Thus $\mathcal{A}^\bullet$ is a fine resolution. Wedge product with a bounded form and bounded weak derivative preserves $\mathcal{L}^\bullet$ and satisfies the Leibniz rule by local regularisation on the regular locus. The resulting pairing extends scalar multiplication by $\mathbb{R}$. Replacing $\mathbb{R}$ by this fine resolution computes the canonical hypercohomology module action [Bre97, II.7]; this proves the asserted compatibility. □

8.2. **The pairing.** Put $m = \dim_{\mathbb{R}} M = 16k + 4$ and let $\mathbb{D}_M$ denote the dualising complex. We use the unsigned intersection pairing $Q : IH^j \times IH^{m-j} \to \mathbb{R}$. Poincaré duality for L^2 cohomology by integration is classical for Witt spaces with iterated conical metrics [Che83b, ALMP18]. For the present metric the identification with the intersection pairing can be proved directly from Stokes and the uniqueness of the self-duality of the intersection complex.

Proposition 8.2. *If closed $\alpha, \beta \in \operatorname{Dom} d_{\max}$ represent $\Phi(a), \Phi(b)$ in complementary degrees, then*

$$Q(a,b) = \int_{M_{\mathrm{reg}}} \alpha \wedge \beta. \tag{40}$$

Proof. The product is integrable by Cauchy–Schwarz. Stokes gives, for $u, v \in \operatorname{Dom} d_{\max}$ with $\deg u = p$ and $\deg u + \deg v = m - 1$,

$$\int du \wedge v + (-1)^p \int u \wedge dv = 0.$$

Indeed approximate u in its graph norm by smooth forms compactly supported in the regular locus and use the weak derivative of v. Thus integration depends only on classes.

For completeness this functional determines the correct sheaf-theoretic pairing, not just an abstract nondegenerate form. Over $\mathbb{R}$, the tensor product $\mathcal{F} = \mathcal{L}^\bullet \otimes \mathcal{L}^\bullet$ represents $\mathcal{IC}_M \otimes^L \mathcal{IC}_M$ and has fine terms. Compact Verdier duality gives

$$\operatorname{Hom}_{D(M)}(\mathcal{F}, \mathbb{D}_M[-m]) \cong \operatorname{Hom}_{\mathbb{R}}(H^m\Gamma(M, \mathcal{F}), \mathbb{R}).$$

Integration of the wedge product defines the right-hand functional. To verify that it kills coboundaries, use a finite partition of unity to express a section as sums of tensors of global sections and apply the displayed Stokes identity. On an open regular subset, the induced functional on compact supports is ordinary integration; hence its morphism restricts to the orientation pairing. Self-duality of $\mathcal{IC}_M$ identifies the space of such pairings with $\operatorname{Hom}(\mathcal{IC}_M, \mathcal{IC}_M)$. An endomorphism of the intersection complex is determined by its restriction to the dense regular locus [GM83, §§3.5 and 5.3]. The morphism is therefore the intersection pairing. Evaluating on $\alpha \otimes \beta$ proves (40). □

In top degree, integration is consequently an isomorphism $H^m_{(2)} \to \mathbb{R}$. It also follows directly from Stokes, $\dim IH^m = 1$ and the positive integral of the top Kähler power. For real harmonic forms the Hodge star preserves $\mathcal{H}$, and

$$\langle \alpha, \beta \rangle_{L^2} = Q(\alpha, *\beta). \tag{41}$$

Moreover $*$ belongs on $\mathcal{H}$ to the associative algebra generated by L_1, Λ_1 and W_1. To see this, the Lefschetz decomposition separates the summands $L_1^r \mathcal{H}^j_{\mathrm{prim}}$. Their projections belong

to the algebra generated by the $\mathfrak{sl}_2$ action, either by its Casimir and h or by the double-centraliser theorem. Weil's star formula sends $L_1^r\eta$ to a scalar multiple of $L_1^{2n-j-r}W_1\eta$ on each such summand [Huy05, Proposition 1.2.31]. Finally W_1 or its inverse is $\exp(\pi K_{23}/2)$ by the pointwise relations. Thus an endomorphism commuting with the LLV algebra commutes with $*$ on $\mathcal{H}$.

8.3. **The positive three-plane.** The forms ω_a are parallel and bounded. They define classes $x_a \in H^2(M;\mathbb{R})$ through the resolution $\mathcal{A}^\bullet$. The restriction to M_{reg} is injective on H^2: terminal factoriality gives $H^2(M) \cong IH^2(M)$, and codimension at least three gives $IH^2(M) \cong H^2(M_{\mathrm{reg}})$ [Tig25, Lemma 2.10, Proposition 2.17]. Recollection 3.1 therefore identifies x_1 as a positive multiple of an ample class and $x_2 + ix_3$ as a nonzero multiple of the symplectic class.

Proposition 8.3. *There is $c > 0$ such that $q_M(x_a, x_b) = c\delta_{ab}$.*

Proof. The module statement and top-degree integration show that evaluation of a product of bounded closed representatives on $[M]$ is a fixed positive multiple of their wedge integral. On a quaternionic tangent space, unit quaternions act by volume-preserving isometries rotating the three Kähler forms. Thus

$$\int (t_1\omega_1 + t_2\omega_2 + t_3\omega_3)^{2n} = C(t_1^2 + t_2^2 + t_3^2)^n, \qquad C > 0.$$

The Fujiki relation [BL22, Proposition 5.15] makes this a positive multiple of $q_M(\sum t_a x_a)^n$. This quadratic form is nonzero away from zero on the three-plane and positive at the ample ray. It is therefore the positive nth root of the displayed polynomial, giving the claim by polarisation. □

Theorem 8.1, Proposition 8.2 and Corollary 7.3 prove all of Theorem 1.2. Section 2 now proves Theorem 1.1 without any remaining analytic premise.

9. Monodromy and the spin representation

The degree-zero part of the LLV algebra contains the rational subalgebra

$$\mathfrak{s} \cong \mathfrak{so}(H^2(M;\mathbb{Q}), q_M).$$

Its representation on $IH^\bullet(M;\mathbb{Q})$ integrates to the simply connected spin group. Let G be its image as a connected rational algebraic group. The bracket formulas of Section 2 give an isogeny

$$G \longrightarrow \mathrm{SO}(H^2(M;\mathbb{Q}), q_M), \qquad \sigma \longmapsto \bar{\sigma},$$

such that

$$\sigma L_x \sigma^{-1} = L_{\bar{\sigma}x}, \qquad \sigma\Lambda_x\sigma^{-1} = \Lambda_{\bar{\sigma}x}, \qquad \sigma h = h\sigma. \tag{42}$$

For a nondegenerate complex subspace $W \subset H^2(M;\mathbb{C})$, write G_W for the connected subgroup of $G_{\mathbb{C}}$ integrating $\mathfrak{so}(W)$, acting trivially on $W^\perp$ in H^2.

Proposition 9.1. *Suppose a group Γ acts on $IH^\bullet(M;\mathbb{Q})$ by graded automorphisms preserving the intersection pairing and a lattice. Suppose also that it acts on $H^2(M;\mathbb{Q})$ by isometries preserving a lattice, and that these actions satisfy $\gamma L_x \gamma^{-1} = L_{\gamma x}$. A subgroup of finite index of Γ acts on IH through $G(\mathbb{Q})$, compatibly with its action on H^2. If the complex Zariski closure of its action on H^2 contains $\mathrm{SO}(W)$, the closure of its action on IH contains G_W.*

Proof. First G preserves the unsigned intersection pairing Q. The module action and Proposition 8.2 give $L_x^\dagger = L_x$ for the Q-adjoint, whereas $h^\dagger = -h$. Taking adjoints of $[L_x, \Lambda_x] = h$ shows that $\Lambda_x^\dagger$ is another degree--2 lowering operator, so $\Lambda_x^\dagger = \Lambda_x$. Hence all brackets $[L_x, \Lambda_y]$ are Q-skew, and so is $\mathfrak{s}$.

Let $G_\mathbb{Z}$ be the stabiliser in $G(\mathbb{Q})$ of the chosen lattice in IH. It is arithmetic; its image under the isogeny is commensurable with the integral orthogonal group [Mor15, (5.4.5), (9.4.5)]. After passing to a subgroup Γ_1 of finite index, every action on H^2 has a lift in $G_\mathbb{Z}$. Consider the group of pairs

$$\widetilde{\Gamma}_1 = \{(\gamma, \sigma) : \gamma \in \Gamma_1,\ \sigma \in G_\mathbb{Z},\ \bar{\sigma} = \gamma|_{H^2}\}.$$

The operator $c(\gamma, \sigma) = \gamma\sigma^{-1}$ preserves the lattice and Q and commutes with all L_x. It commutes with h and therefore with the uniquely determined Λ_x. It centralises the whole LLV algebra and G. In particular c is a homomorphism on the group of pairs.

Transport these operators to harmonic forms. By the star formula after (41), every c commutes with $*$. It preserves the positive inner product $Q(\alpha, *\beta)$. The intersection of a compact orthogonal group with a lattice stabiliser is finite. Thus the image of c is finite, and the image in Γ_1 of $\ker c$ has finite index. Its action on IH is exactly the required action through $G(\mathbb{Q})$.

For the last assertion take the Zariski closure of that finite-index subgroup in $G_\mathbb{C}$. Its projection contains $\mathrm{SO}(W)$ because finite index preserves the identity component of an algebraic closure. The isogeny has finite kernel; its inverse image over $\mathrm{SO}(W)$ has connected component G_W. A closed subgroup projecting onto $\mathrm{SO}(W)$ has the same Lie algebra over this subgroup and contains G_W. □

Apply this proposition to locally trivial monodromy. Parallel transport preserves the integral structures and the module action. It also preserves the intersection pairing: on the regular parts the complex orientations vary continuously in a locally trivial family, so the oriented fundamental classes are a flat section. Naturality of intersection duality then preserves Q. Thus

$$\mathrm{Mon}' \subset G(\mathbb{Q}), \qquad [\mathrm{Mon} : \mathrm{Mon}'] < \infty, \qquad \overline{\mathrm{Mon}}^{\mathrm{Zar}} \supset G_\mathbb{C}. \tag{43}$$

The last inclusion uses the density on H^2 established in Section 2.

The proof is the harmonic-inner-product argument of [Ver13, Theorem 3.5]; for moduli of $K3^{[n]}$ type compare [Mar08, Lemma 4.13]. It applies unchanged to compact smooth hyperkähler manifolds with their ordinary cohomology and Poincaré pairing. This smooth version will be used together with Proposition 9.1 below.

10. The first representation calculations

Write $\mathbb{V}_\lambda$ for the irreducible complex representation of $\mathfrak{so}(25)$ with highest weight λ in the orthogonal coordinates of B_{12}. Cohomology will sometimes be centred by subtracting the complex dimension from its degree. In this convention h is the grading operator.

10.1. **The ten-dimensional comparison.** For $v_0^2 = 2$ and a $2v_0$-generic polarisation, the singular moduli space has the O'Grady symplectic resolution $\widetilde{M}$. This case is outside (1) and requires none of our domain analysis.

Proposition 10.1. *In the ten-dimensional case,*

$$IH^\bullet(M; \mathbb{C}) = \mathbb{V}_{(5)} \oplus \mathbb{V}_{(3)} \oplus \mathbb{V}_{(2,1)} \oplus \mathbb{V}_{(1)}.$$

Proof. The decomposition theorem calculated in [dCRS21] gives

$$H^\bullet(\widetilde{M}) = IH^\bullet(M) \oplus H^\bullet(\operatorname{Sym}^2 M_0)[-2] \oplus H^\bullet(M_0)[-6],$$

compatibly with the $H^\bullet(M)$ action. Each summand has grading symmetric about the middle and is preserved by the raising operators. It is therefore preserved by their lowering operators [FSY22, Lemma 3.4].

The diagonal restriction $H^2(M) \to H^2(M_0)$ is a similitude. It is equivariant for locally trivial monodromy, so its pullback of the Beauville–Bogomolov form is a scalar multiple of q_M by density. This multiple is nonzero on an ample class. Since both spaces have dimension 23, restriction is an isomorphism. Thus the last summand is the LLV module $\mathbb{V}_{(2)}$ of $K3^{[2]}$, and the preceding summand is $\operatorname{Sym}^2 \mathbb{V}_{(2)}$, with the indicated degree shifts.

By [GKLR22], the $\mathfrak{so}(26)$ module of $\widetilde{M}$ is $\mathbb{V}_{(5)} \oplus \mathbb{V}_{(2,2)}$. Restriction to $\mathfrak{so}(25)$, the stabiliser of the exceptional class, is

$$\mathbb{V}_{(5)} \oplus \mathbb{V}_{(4)} \oplus \mathbb{V}_{(3)} \oplus 2\mathbb{V}_{(2)} \oplus \mathbb{V}_{(1)} \oplus \mathbb{V}_{(0)} \oplus \mathbb{V}_{(2,2)} \oplus \mathbb{V}_{(2,1)}.$$

Subtract $\operatorname{Sym}^2 \mathbb{V}_{(2)} = \mathbb{V}_{(4)} \oplus \mathbb{V}_{(2,2)} \oplus \mathbb{V}_{(2)} \oplus \mathbb{V}_{(0)}$ and the last $\mathbb{V}_{(2)}$. Complete reducibility gives the stated decomposition. □

10.2. **A character in dimension eighteen.** Return to a Picard-rank-one quartic S, $v_0 = (5, A, 0)$, and $M = M_A(2v_0)$. Then M_0 is of type $K3^{[3]}$. Centre all gradings. Transport the LLV modules of Hilbert schemes to $\mathfrak{so}(25)$ by complex similitudes of their second cohomologies; their characters are independent of these choices.

Lemma 10.2. *The virtual character* $\operatorname{ch} H^\bullet(K3^{[9]}) - \operatorname{ch} \bigwedge^2 H^\bullet(K3^{[3]})$ *is the character of the module* $\mathcal{R}$ *whose nonzero multiplicities are listed in Table 1. Its dimension is* 138065600.

TABLE 1. The irreducible summands of $\mathcal{R}$. Omitted coordinates of each highest weight are zero.

λ	multiplicity	λ	multiplicity	λ	multiplicity
(9)	1	$(4,2)$	1	$(2,2)$	1
$(7,1)$	1	$(4,1,1)$	1	$(2,1,1)$	1
(7)	1	$(4,1)$	2	$(2,1)$	3
$(6,1)$	1	(4)	2	(2)	3
(6)	1	$(3,2)$	2	$(1,1,1)$	1
$(5,2)$	1	$(3,1)$	4	$(1,1)$	2
$(5,1)$	1	(3)	4	(1)	4
(5)	3	$(2,2,1)$	1		

Proof. Let $z_1, \dots, z_{12}$ be the standard torus coordinates for type B_{12}. The Hilbert-scheme character formula of [GKLR22, Theorem 1.2(1)] is

$$1 + q\sum_{i=1}^{12}(z_i + z_i^{-1}) + \sum_{n\geq 2} \operatorname{ch} H^\bullet(K3^{[n]})q^n = \prod_{m\geq 1}\prod_{i=1}^{12} \frac{1}{(1-q^m z_i)(1-q^m z_i^{-1})}. \tag{44}$$

The convention here centres degrees. The identity is in the formal power series ring over the B_{12} representation ring; its coefficient of q is written separately because $K3^{[1]}$ does not have

the same LLV algebra as $K3^{[n]}$ for $n \geq 2$. Expand through q^9, and use

$$\operatorname{ch} \bigwedge^2 V(z) = \tfrac{1}{2}\big((\operatorname{ch} V(z))^2 - \operatorname{ch} V(z^2)\big).$$

Successive subtraction of highest-weight characters gives the multiplicities in the table. These are finite weight computations: the positive roots are $e_i \pm e_j$ for $i < j$ and e_i, and Freudenthal's recurrence determines each character from its highest weight. The accompanying code performs this calculation with exact integers. At $z = 1$, the result is

$$143184000 - \binom{3200}{2} = 138065600,$$

also the sum of the Weyl dimensions in the table. Replacing the exterior square by the symmetric square gives negative irreducible multiplicities. Repeating the calculation with $K3^{[5]}$ and $K3^{[2]}$ gives Proposition 10.1. □

The character calculation is independent of whether it describes $IH(M)$. To state the geometric inputs separately, let BPS_v be the BPS cohomology of one-dimensional semistable sheaves of vector v on a K3 surface, with the normalisations of [DHS22]. We use the following named hypotheses when indicated:

(H1) On a positive primitive ray $\mathbb{Z}_{>0}w$, the BPS Lie algebra is freely generated by the intersection cohomologies of the moduli spaces on that ray.

(H2) The graded BPS cohomology is independent of the Euler characteristic when the first Chern class is fixed.

These are the BPS and χ-independence statements of the preprints [DHS22, DHKSV26, GWZ26]. Under them, the degree $2w$ part of the free Lie algebra and deformation equivalence [PR23, Theorem 1.7] give a graded-vector-space isomorphism

$$H^\bullet(K3^{[9]}) \cong IH^\bullet(M) \oplus \bigwedge^2 H^\bullet(K3^{[3]}). \tag{45}$$

Indeed the primitive-vector moduli space on the ray has type $K3^{[3]}$; changing the Euler characteristic of the doubled vector to an odd integer makes it primitive and gives type $K3^{[9]}$. The primitive cohomology is even, so its Lie bracket contribution is the ordinary exterior square.

Consequently (H1)–(H2) imply vanishing of odd intersection cohomology and the following Betti numbers through the middle:

(46)

j	0	1	2	3	4
$\dim IH^{2j}$	1	23	300	2876	22427
j	5	6	7	8	9
$\dim IH^{2j}$	150053	887710	4727859	22432662	81617778

The other numbers follow from $\dim IH^{36-2j} = \dim IH^{2j}$. This is still a statement about graded dimensions, not the LLV decomposition.

10.3. **Recovering a representation from a restricted character.** Choose coordinates $e_1, \dots, e_{12}$ on a Cartan subalgebra of $\mathfrak{so}(25)$, with $e_1(h) = 2$ and $e_i(h) = 0$ for $i > 1$. Let $\mathfrak{t}' = \ker e_{12}$.

Lemma 10.3. *Suppose two finite-dimensional $\mathfrak{so}(25)$ modules V, V' have equal characters on $\mathfrak{t}'$. If every highest weight ν of V' has $\nu_{11} = 0$, then $V \cong V'$.*

Proof. Use lexicographic order. Every weight α of $\mathbb{V}_\lambda$ satisfies $\alpha \le \alpha^+ \le \lambda$, where α^+ is obtained by sorting its absolute coordinates. The second inequality follows from the positive root order. The largest restricted highest weights of V and V' therefore have identical first eleven coordinates. If they are λ and ν, the assumption gives $\lambda_{11} = \nu_{11} = 0$, and dominance forces $\lambda_{12} = \nu_{12} = 0$; hence $\lambda = \nu$.

No other weight can contribute to this largest restricted weight. For if α has those first eleven coordinates, then $\alpha \le \alpha^+ \le \lambda' \le \nu$ for its constituent highest weight λ'. The first eleven coordinates of α^+ must agree with ν. Its eleventh and twelfth coordinates are zero, so $\alpha^+ = \lambda' = \nu$ and $\alpha_{12} = 0$. The multiplicity of the restricted weight is therefore exactly the multiplicity of $\mathbb{V}_\nu$. Subtract this common summand and continue by induction. □

The hypothesis is necessary: the sums of $(25-2i)\bigwedge^i \mathbb{C}^{25}$ over even and odd i, for $0 \le i \le 12$, have the same restricted character. Nor do the degree and Hodge gradings alone distinguish representations: $\mathbb{V}_{(1,1,1)} \oplus 230\mathbb{V}_{(1)}$ and $21\mathbb{V}_{(1,1)} \oplus 1750\mathbb{V}_{(0)}$ agree on that two-dimensional torus. These examples also test the scope of the character argument.

10.4. **The Hodge-theoretic hypothesis.** Let (S', β) be a Picard-rank-one K3 surface of degree four, fix $\chi_0 \neq 0$, and put

$$w = (0, \beta, \chi_0), \quad v' = (0, 2\beta, 2\chi_0 + 1), \qquad M' = M_\beta(2w), \quad N' = M_\beta(w), \quad X' = M_\beta(v').$$

The last two spaces have types $K3^{[3]}$ and $K3^{[9]}$, and M' is locally trivially deformation equivalent to M [PR23, Theorem 1.7]. The additional hypothesis is:

(H3) For every such (S', β) there is an isomorphism of graded rational Hodge structures, with usual degrees,

$$H^\bullet(X') \cong IH^\bullet(M') \oplus \bigwedge^2 H^\bullet(N')[-6](-3). \tag{47}$$

Here $[-6]$ raises degrees by six, and (-3) is the Tate twist.

We do not prove (H3). Its proposed source is the mixed-Hodge-module form of the BPS theorem [DHS22, Theorems 9.2 and 10.1, Corollary 3.6, Propositions 6.10 and B.14] and the independence theorem [DHKSV26, Theorem 5.1]. The first splits the BPS object of $2w$ into the intersection object and the antisymmetric direct-sum contribution from $N' \times N'$. The second identifies the perverse direct images under the support maps for different Euler characteristics as mixed Hodge modules. Saito's decomposition theorem [Sai90, (4.5.4)] then gives (47); weights impose the displayed twist. This deduction uses the mixed-Hodge-module assertion of the second theorem, whose maps are constructed by Hecke operators. Equality of Hodge numbers, as supplied by [GWZ26], is insufficient for what follows.

Theorem 10.4. *Assume (H3). For the eighteen-dimensional moduli space in (1), $IH^\bullet(M; \mathbb{C}) \cong \mathcal{R}$ as a module over its LLV algebra.*

Proof. Choose a smooth algebraic family of degree-four polarised K3 surfaces with arithmetic monodromy on $\beta^\perp$ [Huy16, Corollary 6.4.7]. Form the three relative moduli spaces with vectors $2w, w, v'$. Restrict to the open set where the polarisation is *general with respect to* these vectors in the sense of [PR23, Definition 2.8]; the required openness is [PR23, Proposition 2.14]. This is the specific condition used here, and is not an identification with every wall-based genericity convention. The open set contains the Picard-rank-one fibres.

Shrink further to a smooth connected dense open base T° where the singular family is topologically locally trivial with its Whitney strata [Vdr76, Corollaire (5.1)] and the two primitive families are smooth. Intersection cohomology then forms a local system with its

integral structure, pairing and ordinary-cohomology action. In each degree it underlies a polarizable pure variation after a further shrinking. To justify that its fibre structure is the usual intersection Hodge structure, apply Saito's projective direct-image theorem and generic variation theorem [Sai88, Théorèmes 5.3.1 and 5.4.3] to the intersection Hodge module of the total space. Base change [Sai90, (4.4.3)] identifies its restriction to a fibre; local triviality identifies the underlying perverse sheaf with the intersection complex. Its restriction to the regular locus is the constant variation, which determines the intersection extension and its Hodge structure, with the degree shifts and possible Tate normalisation.

Choose a Hodge-generic point of this base whose surface has Picard rank one. Such points exist outside a countable union of proper analytic subsets. Let $\Gamma = \pi_1(T^\circ)$. Removing proper complex subvarieties from a smooth base surjects on its original fundamental group, so the image of Γ on $\beta^\perp$ remains arithmetic and Zariski dense in $\mathrm{SO}(\beta^\perp_{\mathbb{C}})$.

The Mukai isometries for N' and X' identify their second cohomologies with $w^\perp$ and $(v')^\perp$ [Mar08, Theorem 1.7]. These identifications are locally constant [O'G97, Mar08, OPR24]. Their common subspace $\beta^\perp$ has dimension 21, and monodromy is trivial on its two-dimensional complements, consisting of vectors $(r, a\beta, s)$. The diagonal restriction $H^2(M') \to H^2(N')$ is a similitude by the argument in Proposition 10.1. This assertion may be checked using the full locally trivial monodromy of M'; that group also transports its intrinsic diagonal stratum and acts isometrically on $H^2(N')$. Restriction is a morphism of local systems in our chosen family. It identifies a corresponding nondegenerate 21-dimensional subspace $W \subset H^2(M')$ with $\beta^\perp_{\mathbb{C}}$ and identifies their Γ actions.

Transport Proposition 9.1 from M to M'; apply its smooth version to N' and X'. A finite-index subgroup of Γ acts on their cohomologies through the corresponding spin images. All three actions project to the same action on $\beta^\perp$. Their fibre product over $\mathrm{SO}(\beta^\perp_{\mathbb{C}})$ is a finite cover of that group; its identity component is the diagonal image of $\mathrm{Spin}(\beta^\perp_{\mathbb{C}})$. Density of the projection implies that this diagonal image is the connected monodromy group on

$$\mathcal{V} = IH^\bullet(M') \oplus \bigwedge^2 H^\bullet(N') \oplus H^\bullet(X').$$

Degree by degree these are polarizable pure variations; their direct sum is a graded-polarizable variation with several weights. André's monodromy theorem [And92, Lemma 4, Theorem 1] puts its connected monodromy group in the derived group of the generic Mumford–Tate group. It therefore fixes every rational tensor of type (p,p): after a Tate twist such a tensor is Hodge, and the derived group acts trivially on Tate structures. The components of the isomorphism (47), extended by zero to operators on the indicated direct sum, are such tensors. Hence (H3) is equivariant for this diagonal spin group at the chosen generic point. No isomorphism of variations is assumed.

Include h with a Cartan subalgebra of $\mathfrak{so}(W)$. The resulting subalgebra is $\mathfrak{t}' = \ker e_{12}$. Extend the identifications on W to complex similitudes of the full second cohomologies; the complementary quadratic spaces have dimension two, so such extensions exist. After centring the gradings, (47) now says that the restricted character of $IH(M')$ is that of $H(K3^{[9]}) - \bigwedge^2 H(K3^{[3]})$, hence that of $\mathcal{R}$. Every highest weight in Table 1 has at most three nonzero entries. Lemma 10.3 identifies the full modules. Parallel transport from M' to M completes the proof. □

This last argument uses (H3) at a Hodge-generic fibre only. It does not identify Saito's Hodge decomposition with the type decomposition of harmonic forms for the metric on M.

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